



Modelling Financial Markets

Chapter 13: Machine Learning in Finance



Daniel Traian PELE

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material (1/3)

- ▣ ADF: Augmented Dickey–Fuller (test)
- ▣ AI: Artificial Intelligence
- ▣ AIC: Akaike Information Criterion
- ▣ AUC: Area Under the (ROC) Curve
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BH: Benjamini–Hochberg (procedure)
- ▣ CAPM: Capital Asset Pricing Model
- ▣ CI: Confidence Interval
- ▣ CMA: Conservative Minus Aggressive (investment factor)
- ▣ CPCV: Combinatorial Purged Cross-Validation
- ▣ CRSP: Center for Research in Security Prices
- ▣ CSCV: Combinatorially Symmetric Cross-Validation
- ▣ CV: Cross-Validation
- ▣ CW: Clark–West (test for nested models)
- ▣ DM: Diebold–Mariano (test)
- ▣ DML: Double/Debiased Machine Learning
- ▣ DSR: Deflated Sharpe Ratio
- ▣ ELW: Exact Local Whittle (estimator)
- ▣ ES: Expected Shortfall
- ▣ EU: European Union
- ▣ EV: Explained Variation
- ▣ EWMA: Exponentially Weighted Moving Average
- ▣ FDR: False Discovery Rate
- ▣ FF3: Fama–French three-factor model
- ▣ FF5: Fama–French five-factor model
- ▣ FFD: Fixed-width window Fractional Differentiation

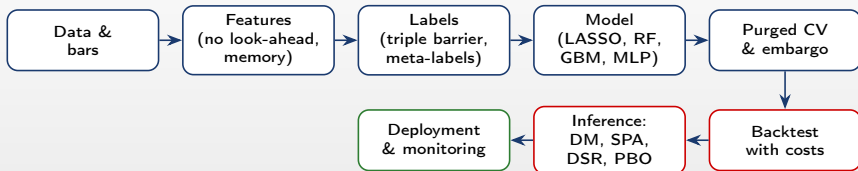
Acronyms used in this material (2/3)

- ▣ **FFN: FeedForward Network**
- ▣ **FN: False Negative**
- ▣ **FRED-MD: Federal Reserve Economic Data, Monthly Database (McCracken–Ng)**
- ▣ **FWER: Family-Wise Error Rate**
- ▣ **GAN: Generative Adversarial Network**
- ▣ **GBM: Gradient Boosting Machine**
- ▣ **GMM: Generalized Method of Moments**
- ▣ **GRS: Gibbons–Ross–Shanken (test)**
- ▣ **GRU: Gated Recurrent Unit**
- ▣ **GW: Giacomini–White (test)**
- ▣ **HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)**
- ▣ **HML: High Minus Low (value factor)**
- ▣ **IBM: International Business Machines (Corporation)**
- ▣ **IS: In-Sample**
- ▣ **LASSO: Least Absolute Shrinkage and Selection Operator**
- ▣ **LIME: Local Interpretable Model-agnostic Explanations**
- ▣ **LLM: Large Language Model**
- ▣ **LS: Linear SDF estimated by unregularised Least Squares**
- ▣ **LSTM: Long Short-Term Memory (network)**
- ▣ **LW: Local Whittle (estimator)**
- ▣ **MA: Moving Average**
- ▣ **MA20: 20-day Moving Average**
- ▣ **MDA: Mean Decrease Accuracy**
- ▣ **MDI: Mean Decrease Impurity**
- ▣ **ML: Machine Learning**
- ▣ **MLP: MultiLayer Perceptron**

Acronyms used in this material (3/3)

- ▣ MSCA: Marie Skłodowska-Curie Actions
- ▣ OLS: Ordinary Least Squares
- ▣ OOS: Out-Of-Sample
- ▣ PAC: Probably Approximately Correct (learning)
- ▣ PBO: Probability of Backtest Overfitting
- ▣ PCR: Principal Component Regression
- ▣ PLS: Partial Least Squares
- ▣ PSR: Probabilistic Sharpe Ratio
- ▣ PT: Pesaran–Timmermann (test of directional accuracy)
- ▣ RC: Reality Check (White's test for data snooping)
- ▣ RF: Random Forest
- ▣ RL: Reinforcement Learning
- ▣ RMW: Robust Minus Weak (profitability factor)
- ▣ ROC: Receiver Operating Characteristic (curve)
- ▣ RSI: Relative Strength Index
- ▣ SD: Standard Deviation
- ▣ SDF: Stochastic Discount Factor
- ▣ SE: Standard Error
- ▣ SHAP: SHapley Additive exPlanations
- ▣ SPA: Superior Predictive Ability (test)
- ▣ SR: Sharpe Ratio
- ▣ TN: True Negative
- ▣ TP: True Positive
- ▣ US: United States
- ▣ VIX: CBOE Volatility IndeX
- ▣ WF: Walk-Forward (validation)

Chapter Route: the Financial ML Workflow



- ▣ Part I: the prediction problem, a minimal toolbox, features and fractional differencing, labels, honest validation
- ▣ Part II: an S&P 500 case study with formal tests, selection and backtest overfitting, machine learning (ML) for inference, a research case study, AI for scientific discovery; technical detail sits in the [Appendix](#)
- ▣ After this chapter you can: build features and labels without look-ahead; validate with purging and embargo; test a classifier and a backtest against a benchmark (PT, DM, DSR, SPA, PBO)
- ▣ **Look-ahead bias**: using information not available at decision time; **selection bias**: reporting the best of many trials; [López de Prado \(2018\)](#): most failures come from the process, not the algorithm

Why Machine Learning in Finance, and Why It Is Hard

The promise

- Flexible, non-linear functional forms; interactions found automatically
- Hundreds of candidate predictors (the “factor zoo”, [Chapter 3](#))
- Alternative data: text, images, order books, on-chain data

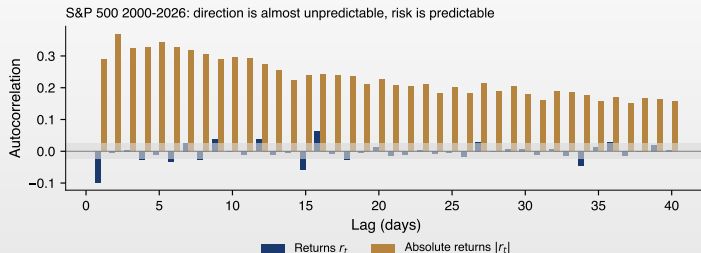
The constraints

- Very low **signal-to-noise ratio**; one serially dependent history that cannot be re-run
- **Non-stationarity**: regimes, breaks; markets react to their users
- Labels must be constructed; errors cost real money
- Massive risk of **overfitting** (fitting noise)

Adaptive Markets Hypothesis: Lo (2004)

- Predictability appears, is arbitrated away and re-appears: a model that worked yesterday is a **hypothesis**, not a law (market efficiency: [Chapter 2](#))
- The hard part is not fitting a model but **not fooling yourself**: correct labels, correct validation, honest backtests

What Is Predictable? Direction vs. Risk



- Autocorrelation at lag k : $\rho_k = \text{Corr}(x_t, x_{t-k})$, the correlation of a series with its own value k days earlier
 - x_t : the daily log return r_t or its absolute value $|r_t|$; T : number of observations
 - grey band: $\pm 1.96/\sqrt{T}$, the 95% band under no autocorrelation; a bar outside it is significant at 5%
- Autocorrelations of returns r_t are tiny (inside the band); those of $|r_t|$ are large and persistent
- ML has the best chance where there is structure: **volatility, risk, liquidity, cross-sectional ranking**

Evidence, and the Economic Value of a Small R^2_{OOS} (1/2)

- ▣ [Gu, Kelly & Xiu \(2020\)](#): monthly returns of individual US stocks; linear models, penalised regressions, principal component regression (PCR), partial least squares (PLS), forests, boosted trees, neural nets
- ▣ Their fit measure, the out-of-sample R^2 : how much of the squared return a forecast removes, relative to forecasting zero

$$R^2_{OOS} = 1 - \frac{\sum_t (r_t - \hat{r}_t)^2}{\sum_t r_t^2}$$

- ▶ r_t : realised return in month t of the test period; $\hat{r}_t$: forecast made with information up to $t - 1$
- ▶ the denominator is the error of the zero forecast $\hat{r}_t = 0$
- ▶ $R^2_{OOS} > 0$: the model beats the zero forecast; $R^2_{OOS} < 0$: it does worse; at most 1 (perfect forecasts)
- ▣ Result: R^2_{OOS} is small (well below 1% a month); trees and nets win through **non-linear interactions**; momentum, liquidity and volatility dominate
- ▣ The benchmark matters: [Welch & Goyal \(2008\)](#) show that most published predictors lose to the **historical mean**; [Gu, Kelly & Xiu \(2020\)](#) and [Chapter 14](#) use a zero forecast

Evidence, and the Economic Value of a Small R^2_{OOS} (2/2)

- ▣ Campbell & Thompson (2008): what a small R^2 is worth to an investor; returns have a predictable and an unpredictable part

$$r_{t+1} = \mu + x_t + \varepsilon_{t+1}, \quad R^2 = \frac{\text{Var}(x_t)}{\text{Var}(r_{t+1})}$$

- ▶ μ : mean return; x_t : predictable component, known at t , mean zero; ε_{t+1} : unpredictable shock
- ▶ R^2 : share of the return variance explained by the predictor
- ▣ An unconstrained mean-variance investor who knows x_t (no costs, no estimation error) reaches the Sharpe ratio SR^*

$$SR^{*2} = \frac{SR_0^2 + R^2}{1 - R^2}$$

- ▶ Sharpe ratio (SR): mean excess return divided by its standard deviation
- ▶ SR_0 : SR of buy-and-hold, which ignores x_t ; $R^2 = 0$ gives $SR^* = SR_0$
- ▣ S&P 500, monthly log returns 2001–2026: $SR_0 = 0.128$ a month (0.44 annualised); $R^2 = 0.5\%$ a month $\Rightarrow SR^* = 0.51$ (+15%), $R^2 = 1\% \Rightarrow 0.57$ (+27%)
- ▣ For a **single index** the effective sample is far smaller than for thousands of stocks: the S&P 500 case study tests exactly this

Loss, Penalty and Five Model Families (1/3)

- Supervised learning picks, within a model family, the function that fits the training data best after a penalty for complexity

$$\hat{f} = \arg \min_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n L(y_i, f(x_i)) + \lambda \Omega(f)$$

- Notation

- ▶ (y_i, x_i) , $i = 1, \dots, n$: target and feature vector of training observation i ; $\mathcal{F}$: the model family (e.g. all trees of depth 3)
- ▶ L : loss, the squared error $(y - f)^2$ or, for a probability f of $y = 1$, the log-loss $-[y \log f + (1 - y) \log(1 - f)]$
- ▶ $\Omega(f)$: complexity penalty; $\lambda \geq 0$: its weight; $\lambda = 0$: no penalty, large λ : a simpler model
- ▶ λ is chosen by K -fold cross-validation (CV): split the data into K folds, fit on $K - 1$, score on the fold left out, average the K scores
- ▶ Expected test error = bias² + variance + noise σ^2 (Hastie, Tibshirani & Friedman (2009), Ch. 7)
 - ▶ bias: error of a too-simple family; variance: sensitivity of $\hat{f}$ to the training sample; σ^2 : variance of the unpredictable part of y
 - ▶ in finance σ^2 dominates, so variance control matters more than flexibility

Loss, Penalty and Five Model Families (2/3)

- Penalised logit / least squares: the penalty is a mix of absolute and squared coefficients

$$\lambda \Omega(\beta) = \lambda \left[\alpha \|\beta\|_1 + \frac{1-\alpha}{2} \|\beta\|_2^2 \right], \quad \|\beta\|_1 = \sum_j |\beta_j|, \quad \|\beta\|_2^2 = \sum_j \beta_j^2$$

- ▶ $\beta = (\beta_1, \dots, \beta_p)$: coefficients of the p features; $\alpha \in [0, 1]$: share of the absolute-value penalty
 - ▶ Ridge, $\alpha = 0$ (Hoerl & Kennard (1970)): shrinks all coefficients towards zero, none exactly to zero
 - ▶ LASSO, $\alpha = 1$ (Tibshirani (1996)): sets some coefficients exactly to zero, i.e. selects features
 - ▶ Elastic Net, $0 < \alpha < 1$ (Zou & Hastie (2005)): in between
- The linear benchmark is the logit, the probability of “up” as a function of a linear score

$$p = P(y = 1 \mid x) = \frac{1}{1 + e^{-x^\top \beta}}$$

- ▶ $x^\top \beta = \sum_j \beta_j x_j$: linear score; p rises from 0 to 1 as the score rises
- ▶ $\beta_j > 0$: a higher x_j raises the probability of “up”

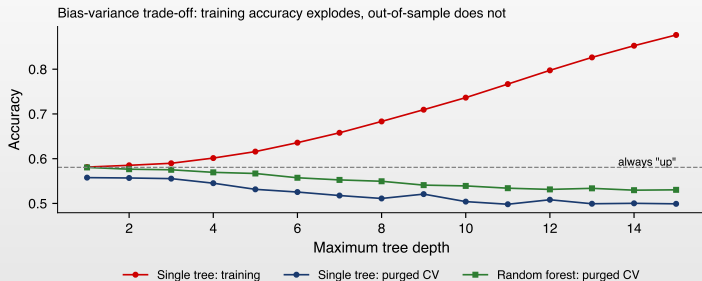
Loss, Penalty and Five Model Families (3/3)

- Trees: recursive splits that minimise the Gini impurity $G = 1 - \sum_c \hat{\pi}_c^2$ of the resulting nodes
 - ▶ $\hat{\pi}_c$: share of class c in a node; $G = 0$: pure node; $G = 0.5$: two classes 50/50
- Random forest (RF): average of B trees, each grown on a bootstrap sample with a random subset of features at each split, so the trees are decorrelated ([Breiman \(2001\)](#)); its out-of-bag error is **optimistic** when labels overlap
- Gradient boosting machine (GBM, [Friedman \(2001\)](#)): $F_m = F_{m-1} + \eta h_m$
 - ▶ F_m : the model after m steps; h_m : a small tree fitted to the negative gradient of the loss (for squared error, the residuals)
 - ▶ $\eta \in (0, 1]$: learning rate; a small η learns slowly and overfits less
- Multilayer perceptron (MLP): $h^{(\ell)} = g(W^{(\ell)}h^{(\ell-1)} + b^{(\ell)})$, $h^{(0)} = x$
 - ▶ $h^{(\ell)}$: output of layer ℓ ; $W^{(\ell)}$, $b^{(\ell)}$: weight matrix and bias vector of layer ℓ
 - ▶ g : ReLU, $g(z) = \max(0, z)$, element by element; early stopping: stop training when the validation loss stops falling; architectures in [Chapter 14](#)
- This chapter classifies the 5-day direction; every CV score is an estimate and is tested against a benchmark



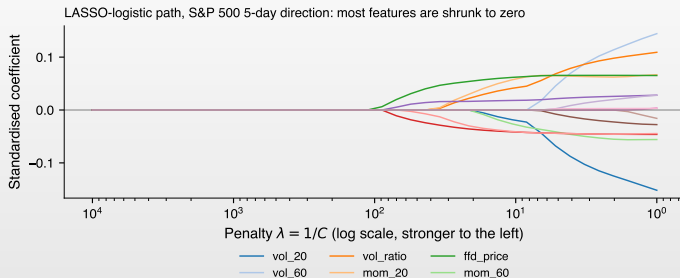
Random Forests

Bias–Variance in Practice: Tree Depth



- S&P 500, 5-day direction; accuracy = share of correctly classified days
- Training accuracy rises to $\approx 87\%$ with depth; purged-CV accuracy **falls** towards 50%
- Random forests degrade more slowly, but never beat the “always up” baseline (dashed line)

LASSO Regularisation Path



- L_1 -penalised logistic regression for the 5-day direction of the S&P 500, 14 standardised features
- Strong penalty: every coefficient is zero; the first feature to enter is the FFD price, a scaled price level (next section)
- Weak penalty: the volatility features (20- and 60-day volatility and their ratio) get the largest coefficients
- All coefficients are small in absolute value: weak signal, consistent with near-efficient markets

Stationarity, Memory and the Fractional Difference (1/2)

- ▣ ML inputs need a distribution that does not drift
 - ▶ log prices $X_t = \log P_t$ (P_t : closing price on day t) behave as $I(1)$
 - ▶ returns $\Delta \log P_t = \log P_t - \log P_{t-1}$ are $I(0)$ but **erase the memory** of the price level
- ▣ Integration order: a series X_t is $I(\delta)$ if $(1 - B)^\delta X_t$ is $I(0)$
 - ▶ B : backshift operator, $BX_t = X_{t-1}$, so $(1 - B)X_t = X_t - X_{t-1}$ is the ordinary difference
 - ▶ $I(0)$: stationary with short memory; $I(1)$: a random walk type series, stationary after one difference
 - ▶ δ may be fractional: the series is stationary only for $\delta < \frac{1}{2}$ ([Hosking \(1981\)](#)); $0 < \delta < \frac{1}{2}$: stationary with long memory
- ▣ Idea of fractional differencing: difference by less than 1, removing the unit root while keeping part of the memory

Stationarity, Memory and the Fractional Difference (2/2)

Definition 1 (Fractional difference of order d)

$$\boxed{\tilde{X}_t = (1 - B)^d X_t = \sum_{k=0}^{\infty} w_k X_{t-k}, \text{ with } w_0 = 1 \text{ and } w_k = -w_{k-1} \frac{d-k+1}{k}}$$

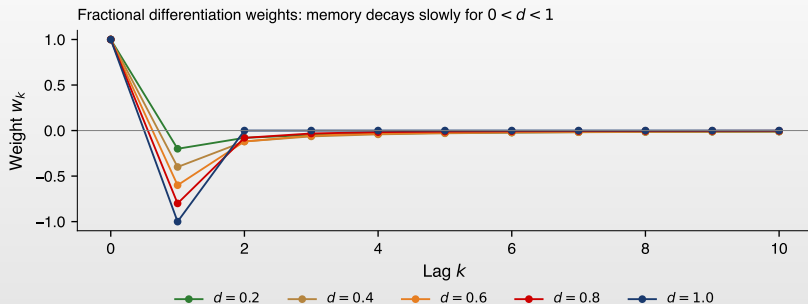
- ▣ d : differencing order chosen by the analyst; k : lag in days; w_k : weight of the value k days back, from the binomial expansion of $(1 - B)^d$
- ▣ $d = 1$: $w = (1, -1, 0, \dots)$, ordinary returns; $0 < d < 1$: weights decay hyperbolically, so memory is kept
- ▣ An $I(1)$ log price differenced with d is $I(1 - d)$: non-stationary for $d \leq \frac{1}{2}$, stationary with long memory for $\frac{1}{2} < d < 1$
- ▣ The infinite weights sum to $(1 - 1)^d = 0$: the level cancels only in the limit (asymptotics in the [Appendix](#))
- ▣ [López de Prado \(2018\)](#) proposes the minimum d at which an ADF test rejects; this section shows why that rule fails

The Fixed-Width Window (FFD)

Definition 2 (Fixed-width-window fractional differentiation (FFD))

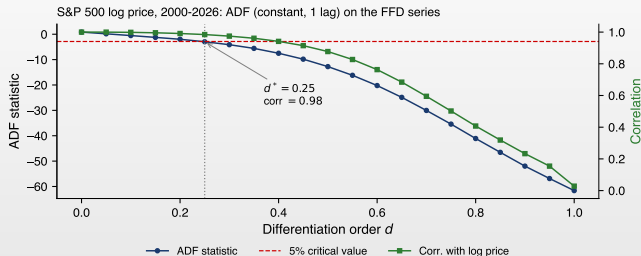
- ▣ Truncate the weights: keep $w_0, \dots, w_K$ with $|w_k| \geq \tau$ (e.g. $\tau = 10^{-4}$), drop the rest (López de Prado (2018))
- ▣ $\tilde{X}_t = \sum_{k=0}^K w_k X_{t-k}$ for every $t > K$
- ▣ τ : smallest weight kept; K : the last lag kept, i.e. the window length; the first K days have too little history and are dropped
- ▣ Every $\tilde{X}_t$ uses the same number of lags; only past values enter $\Rightarrow$ no look-ahead
- ▣ The smaller d , the more lags are needed: $d = 0.25, \tau = 10^{-4} \Rightarrow K = 444$ lags
- ▣ The truncated weights no longer sum to zero: $\sum_{k=0}^K w_k = 0.178$ at $d = 0.25$ (0.040 at $d = 0.5$)

Fractional Differentiation Weights



- For $d = 1$ only the first lag matters; for small d the weights of distant lags remain non-zero
- All weights after w_0 are negative: $\tilde{X}_t$ compares today's level with a weighted history

The ADF Statistic Along d : S&P 500



- Augmented Dickey–Fuller (ADF) unit-root test (Time Series course): the t -statistic of γ in $\Delta \tilde{X}_t = c + \gamma \tilde{X}_{t-1} + (\text{lagged } \Delta \tilde{X}) + e_t$
 - $H_0: \gamma = 0$, a unit root; more negative values are stronger evidence of stationarity; reject below ≈ -2.86 at 5%
- FFD log price, S&P 500 2000–2026, constant and 1 lag: the ADF rejects from $d = 0.25$, where $\text{Corr}(\tilde{X}_t, \log P_t) = 0.98$, the correlation of the FFD series with the log price
- Same series, lag length chosen by AIC (34 lags): ADF = -0.72 , no rejection
- A correlation of 0.98 with an $I(1)$ price is a warning, not a virtue

Truncated FFD of an $I(1)$ Price Is Still $I(1)$

- Decompose the FFD series:

$$\tilde{X}_t = \sum_{k=0}^K w_k X_{t-k} = \left(\sum_{k=0}^K w_k \right) X_t - \sum_{k=1}^K w_k (X_t - X_{t-k})$$

- ▶ first term: today's level times the sum of the truncated weights; second term: a weighted sum of the k -day changes $X_t - X_{t-k}$
- If X_t is $I(1)$, each $X_t - X_{t-k}$ is $I(0)$; the first term is $0.178 X_t$ at $d = 0.25$: a **scaled random walk**
- The unit-root component has small variance next to the stationary part, so ADF with few lags confuses the two
 - ▶ Monte Carlo: pure random walk, $T = 6,274$ days, FFD with $d = 0.25$: ADF (1 lag) rejects in **97.8%** of 500 samples, at a nominal 5%
- Consequence: the FFD log-price in the case study is a scaled price level, not a stationary feature; the next slides measure the memory directly

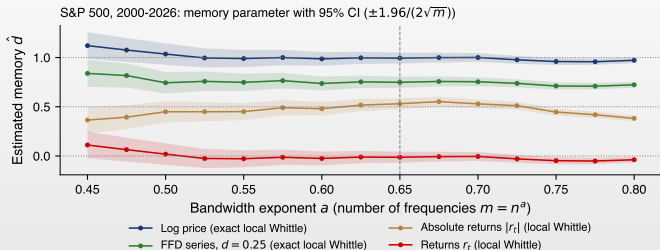
Memory Estimated Directly: Local Whittle (1/2)

- Idea: a long-memory series has most of its variance in slow cycles; the slope of its spectrum near frequency 0 reveals δ

$$f(\lambda) \sim G \lambda^{-2\delta} \quad \text{as } \lambda \rightarrow 0$$

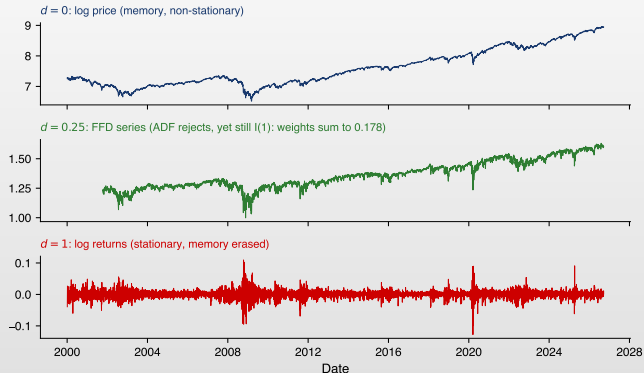
- ▶ $f(\lambda)$: spectral density, the variance contributed by cycles of frequency λ (period $2\pi/\lambda$ days)
- ▶ $G > 0$: scale constant; $\sim$: the ratio of the two sides tends to 1
- ▶ $\delta > 0$: f explodes near 0 (long memory); $\delta = 0$: flat (short memory); $\delta < 0$: f vanishes near 0
- Local Whittle (LW, [Robinson \(1995\)](#)): fits this law only on the first $m = n^a$ Fourier frequencies $\lambda_j = 2\pi j/n$, $j = 1, \dots, m$
 - ▶ n : sample size; $a \in (0, 1)$: bandwidth exponent; a larger a uses more frequencies: less variance, more bias from short-run dynamics
- Exact local Whittle (ELW, [Shimotsu & Phillips \(2005\)](#)): also valid for non-stationary $\delta \geq \frac{1}{2}$
- 95% confidence interval (CI): $\hat{\delta} \pm 1.96/(2\sqrt{m})$; the standard error $1/(2\sqrt{m})$ depends only on m

Memory Estimated Directly: Local Whittle (2/2)



- S&P 500, 2000–2026, $m = n^{0.65}$: log price (ELW) 0.99 [0.94; 1.05]; FFD series (ELW) 0.75 [0.69; 0.81]; returns -0.01 $[-0.07; 0.04]$; $|r_t|$ 0.53 [0.47; 0.59]
- The FFD series stays near $1 - d$ over the band (its order is still 1); the long memory is in **volatility** (Chapters 5 and 9): fractional features belong on $|r_t|$ (Franke, Härdle & Hafner (2019))

Prices, FFD Series and Returns



- ▣ **Top:** log price — $I(1)$
- ▣ **Middle:** FFD with $d = 0.25$ — the upward trend of the price is still visible: $0.178 \log P_t$ plus a stationary part
- ▣ **Bottom:** log returns — only volatility clusters are visible
- ▣ The FFD series enters the case study as the feature “FFD log-price”

From Fixed-Horizon to Triple-Barrier Labels (1/2)

- ▣ **Fixed horizon:** the label is the sign of the h -day log return, with a dead zone around zero

$$r_{t,t+h} = \log P_{t+h} - \log P_t, \quad y_t = \begin{cases} +1 & \text{if } r_{t,t+h} > \tau \\ -1 & \text{if } r_{t,t+h} < -\tau \\ 0 & \text{otherwise} \end{cases}$$

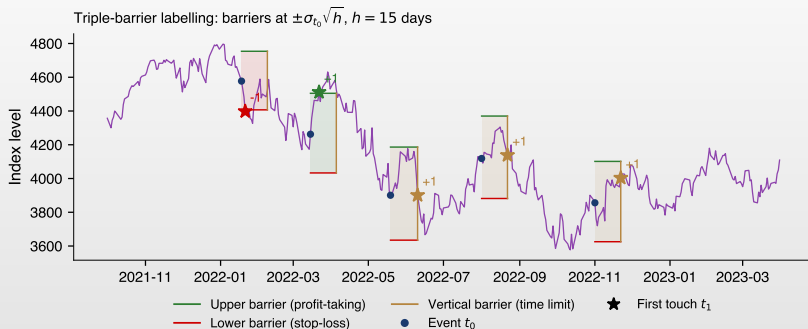
- ▶ P_t : closing price on day t ; h : horizon in days; $\tau \geq 0$: threshold, moves smaller than τ get the label 0
- ▣ Three problems
 - ▶ a constant τ ignores time-varying volatility
 - ▶ the path is ignored: a stop-loss could have been hit before $t + h$
 - ▶ the classes are unbalanced: S&P 500, $h = 10$, $\tau = 0$: 60.5% up vs. 39.5% down

From Fixed-Horizon to Triple-Barrier Labels (2/2)

Definition 3 (Triple-barrier label, López de Prado (2018))

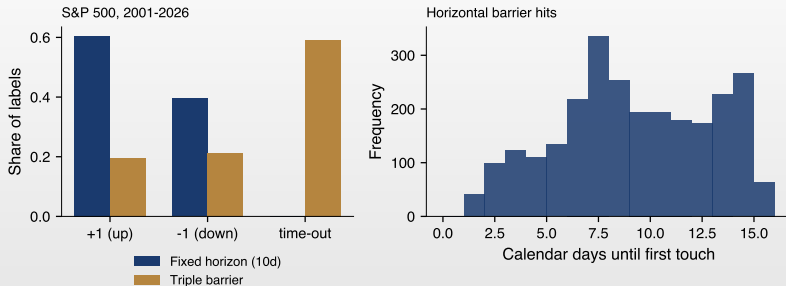
- ▣ At event time t_0 , with daily volatility σ_{t_0} and horizon h : upper barrier $\log P_t - \log P_{t_0} \geq \pi_u \sigma_{t_0} \sqrt{h}$ (profit-taking), lower barrier $\leq -\pi_l \sigma_{t_0} \sqrt{h}$ (stop-loss), vertical barrier $t = t_0 + h$
- ▣ Label = **first barrier touched** at t_1 : +1 upper, -1 lower; at the vertical barrier $\text{sign}(r_{t_0, t_1})$ (or 0)
- ▣ Notation
 - ▶ $\sigma_{t_0} \sqrt{h}$: the typical h -day move; $\pi_u, \pi_l > 0$: barrier widths in units of it
 - ▶ t_1 : day the first barrier is touched; $\text{sign}(x) = +1$ if $x > 0$, -1 if $x < 0$
- ▣ σ_{t_0} : square root of the exponentially weighted moving average (EWMA) of squared deviations of daily log returns, span 50
 - ▶ decay factor $\lambda = 1 - 2/51$: each day weighs λ times as much as the following day
 - ▶ the barriers scale with volatility, like a trader's take-profit, stop-loss and maximum holding period
- ▣ Each label has a **span** $[t_0, t_1]$: the input of purging in validation

Triple Barrier: S&P 500 in 2022



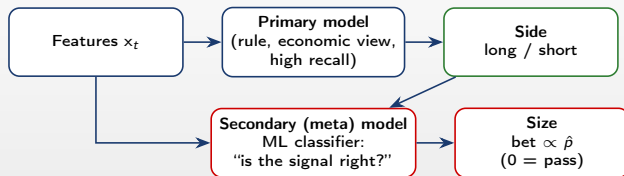
- Five events during the 2022 bear market, $\pi_u = \pi_l = 1$, $h = 15$ days
- January: stop-loss hit (-1); March: profit-taking ($+1$); the others end at the vertical barrier

Label Distribution: Fixed Horizon vs. Triple Barrier



- S&P 500, 2001–2026, $h = 10$: fixed horizon gives 60.5% up / 39.5% down
- Triple barrier ($\pm 1\sigma\sqrt{h}$): 19.6% up, 21.3% down, 59.0% time-out — balanced directional labels
- Right panel: horizontal barriers are typically hit after 1–2 weeks

Meta-Labeling: Deciding Whether to Act (1/2)



- ▣ Meta-label $y_t^{meta} = 1$ if the primary signal was profitable (barrier touched in its direction), 0 otherwise
- ▣ The secondary model estimates $\hat{p} = P(y_t^{meta} = 1 \mid x_t)$, the probability that the signal of day t is right given the features x_t
- ▣ ML learns **when to trust** an interpretable primary rule, not the direction itself

Meta-Labeling: Deciding Whether to Act (2/2)

- Three scores of a filter that decides which signals to act on
 - ▶ precision: share of the trades taken that were profitable
 - ▶ recall: share of the profitable signals that were taken; acting on every signal gives recall 1
 - ▶ $F_1 = 2 \cdot \text{precision} \cdot \text{recall} / (\text{precision} + \text{recall})$: their harmonic mean, between 0 and 1
- Meta-labelling can raise precision at the cost of recall
- Seminar B9, Bitcoin MA20/MA50 crossover, 1,660 test trades: precision 0.475 $\rightarrow$ 0.517, recall 1 $\rightarrow$ 0.541, F_1 0.644 $\rightarrow$ 0.529; gross return per trade $-0.27\% \rightarrow +0.19\%$
- Filtering need not improve F_1 ; with costs per trade, precision and the return per trade are what matter

Overlapping Labels and Sample Uniqueness (1/2)

- Label i spans $[t_{0,i}, t_{1,i}]$: from its start $t_{0,i}$ to its end $t_{1,i}$; consecutive labels share returns $\Rightarrow$ samples are **not i.i.d.**
- Concurrency: the number of labels whose span contains day t

$$c_t = \sum_i 1\{t_{0,i} \leq t \leq t_{1,i}\}$$

- ▶ $1\{A\}$: indicator, 1 if the condition A holds and 0 otherwise
- Average uniqueness of label i : the average share of each day of its span that belongs to it alone

$$\bar{u}_i = \frac{1}{t_{1,i} - t_{0,i} + 1} \sum_{t=t_{0,i}}^{t_{1,i}} \frac{1}{c_t}$$

- ▶ $\bar{u}_i = 1$: no shared return; $\bar{u}_i = 1/5$: the harmonic mean of c_t over the span of label i is five

Overlapping Labels and Sample Uniqueness (2/2)

Consequences

- Effective sample size $\approx \sum_i \bar{u}_i \ll n$, with n the number of labels (an overlap heuristic, not a variance-based measure)
- Bagging draws near-duplicates; CV scores too optimistic

Remedies

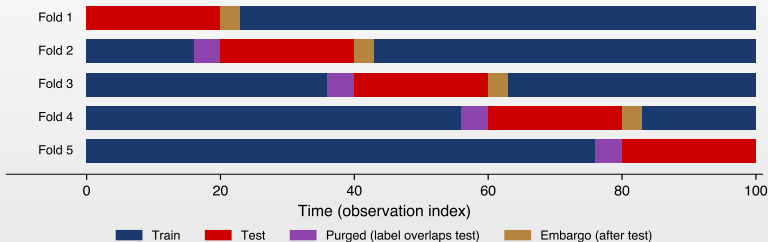
- Weight samples by $\bar{u}_i$ (and by $|r|$)
- Sequential bootstrap; in RF, the fraction of observations drawn for each tree $\approx \bar{u}$, the mean uniqueness

Leakage: What Crosses the Train/Test Boundary

Definition 4 (Leakage)

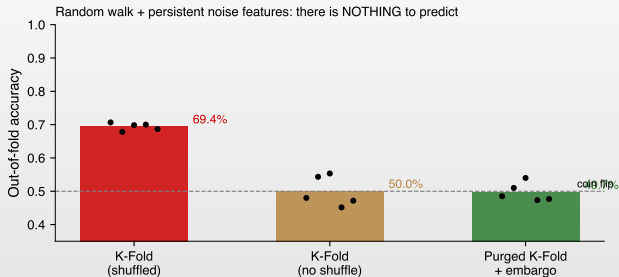
- The training set contains information that is also present in the test set, so the test score no longer measures performance on genuinely new data
- Serially correlated features ($x_t \approx x_{t+1}$) and overlapping labels ($y_t \approx y_{t+1}$): a shuffled K -fold puts **near-identical** days on both sides
- A flexible model (RF, GBM) memorises the neighbour $\Rightarrow$ excellent “out-of-sample” score; tuning on a leaky CV selects the **most overfitted** model
- Other sources: transformations fitted on the full sample (scaling, principal component analysis, the choice of d for FFD), survivorship bias, data revisions, wrong timestamps

Purged K-Fold with Embargo



- ❑ Folds are **contiguous** blocks in time (López de Prado (2018)); **purging** removes every training observation whose label span $[t_{0,i}, t_{1,i}]$ overlaps the span of the test labels, from the first test t_0 to the last test t_1
- ❑ **Embargo**: also remove a small share (e.g. 1%) of observations right after the test block, because features look back in time
- ❑ One boundary: labels end 2 days after they start, test $\{4, 5\}$: observations 2 and 3 end inside the test block (purged), 6 is the embargo, 7 starts before the last test label ends (purged); full table in the [Appendix](#)

Leakage Experiment: Nothing to Predict



- Price = pure random walk; label = sign of the 20-day forward return; 5 features = 20-day rolling means of independent noise
- True accuracy is 50% by construction; random forest, 5 folds
- Shuffled K-fold: 69.4%**; contiguous K-fold 50.0%; purged K-fold + embargo 49.7%
- Shuffling fails because labels overlap *and* features persist; a result that looks too good is usually leakage

 MFM_ch13_purged_cv

Transformations on Training Data Only: the FFD Ablation

- The “minimum d ” was chosen on 2000–2026 and then used inside purged CV: the test folds helped choose a feature
 - ▶ the same ADF rule on data up to 2009 gives $d = 0.15$; on the full sample $d = 0.25$
- Case study without the FFD log-price (purged 5-fold, mean over folds)
 - ▶ accuracy: logit 57.5%, RF 56.4%, GBM 55.6%, MLP 55.5%, all below 58.1%
 - ▶ AUC: 0.483, 0.522, 0.514, 0.514
- The negative conclusion does not depend on this feature
- Correct protocol
 - ▶ estimate d (local Whittle, with its CI) inside each training fold only
 - ▶ prefer features with an economic meaning: distances to moving averages, fractional filters of $|r_t|$

Model Selection versus Deployment

- Ordinary K -fold is not always invalid: for a purely autoregressive model with serially uncorrelated errors it is valid (Bergmeir, Hyndman & Koo (2018))
 - ▶ it fails when the errors are serially correlated: overlapping labels, persistent features outside the model's lags; check the autocorrelation of the CV residuals first
- **Model selection:** purged K -fold, or combinatorial purged CV (CPCV), which tests on many combinations of groups and returns a **distribution** of Sharpe ratios instead of one number
- **Deployment:** walk-forward (WF), train on $[0, T_k]$, test on $(T_k, T_{k+1}]$, roll forward (T_k : end of the k -th training window); realistic, but a single path with high variance
- Rule of thumb: purged K -fold or CPCV to choose, walk-forward for the final, realistic evaluation (details in the [Appendix](#))

Case Study: Data, Target and Validation

- ▣ S&P 500 index and VIX, daily closes, 2001–2026: $n = 6,269$ observations
- ▣ Target: $y_t = 1\{\log P_{t+5} - \log P_t > 0\}$, equal to 1 if the index is higher 5 days later; label span $[t, t + 5]$ used for purging
- ▣ Validation: purged 5-fold CV, 1% embargo; every model gets the same folds
- ▣ Metrics, against the “always up” baseline
 - ▶ accuracy: share of days whose direction is predicted correctly
 - ▶ receiver operating characteristic (ROC) area under the curve (AUC): the probability that a randomly chosen up day gets a higher predicted probability than a randomly chosen down day; 0.5: no skill, 1: perfect ranking
 - ▶ **fold mean**: the metric computed in each of the 5 test folds, then averaged
 - ▶ **pooled**: the metric computed once on all out-of-fold predictions together; used by the tests
- ▣ Chapter data: daily S&P 500 and VIX closes of the course data

Case Study: 14 Features and Four Models

Features, information up to day t

- **Momentum:** $\log P_t - \log P_{t-h}$, the log return over the last h days, $h \in \{1, 5, 20, 60, 120\}$
- **Realised volatility:** annualised standard deviation (SD) of daily log returns over 20 and 60 days, and their ratio
- **Distance to moving average:** $\log P_t$ minus the mean of $\log P$ over 50 or 200 days
- **RSI(14):** $100 - 100/(1 + RS)$, RS = smoothed average gain over average loss, 14 days; between 0 and 100, high after rises
- **VIX:** log level and 5-day change
- **FFD** log price, $d = 0.25$, chosen on the full sample (ablation in the validation section)

Models, deliberately conservative

- **Logit:** L2-penalised (Ridge), standardised features
- **RF:** 400 trees, minimum leaf 100
- **GBM:** histogram-based, depth 3, learning rate 0.03, 200 trees
- **MLP:** hidden layers of 32 and 16 units, early stopping
- Large leaves and shallow trees limit variance

The Model That Learned “Always Up” (1/2)

	Pred. up	Pred. down
Actual up	TP = 3,572	FN = 68
Actual down	FP = 2,591	TN = 38

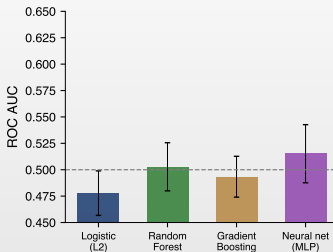
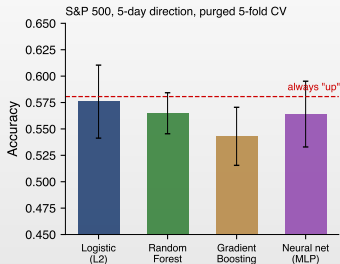
- L2-penalised logit, pooled out-of-fold predictions of purged 5-fold CV, $n = 6,269$
- TP, FN, FP, TN: true positives, false negatives, false positives, true negatives

- Accuracy = $(TP + TN)/n = 57.6\%$: share of days classified correctly
- Precision = $TP/(TP + FP) = 0.580$: share of predicted up days that were up
- Recall = $TP/(TP + FN) = 0.981$: share of actual up days predicted up
- $F_1 = 2 \cdot \text{precision} \cdot \text{recall}/(\text{precision} + \text{recall}) = 0.729$: their harmonic mean
- The model predicts “up” on 98.3% of days: it has essentially learned “always up”

The Model That Learned “Always Up” (2/2)

- ▣ **Majority baseline** from the same table: actual up days $(3,572 + 68)/6,269 = 3,640/6,269 = 58.1\%$
 - ▶ the accuracy of predicting “up” every day, with no model at all
- ▣ A 57.6%-accurate model is *worse* than doing nothing
- ▣ Other baselines: yesterday’s direction, a moving-average rule, buy-and-hold for trading
- ▣ A model has value only if it beats the baseline **out of sample**, after costs, by more than sampling error

Model Comparison, Fold Means



Fold mean	Logit	RF	GBM	MLP
Accuracy	57.6%	56.5%	54.3%	56.4%
AUC	0.478	0.503	0.493	0.515

- Bars: mean over the 5 purged folds; whiskers: ± 1 SD across folds
- Baseline “always up”: **58.1%**; no model beats it

Are the Differences Significant? Pesaran–Timmermann and AUC Intervals (1/2)

- **Pesaran–Timmermann** (PT, Pesaran & Timmermann (1992)): are predicted and realised directions related?

$$PT = \frac{\hat{P} - \hat{P}^*}{\widehat{SE}} \xrightarrow{d} N(0, 1), \quad \hat{P}^* = \hat{p}_y \hat{p}_x + (1 - \hat{p}_y)(1 - \hat{p}_x)$$

- ▶ $\hat{P}$: share of days with the correct sign; $\hat{p}_y$, $\hat{p}_x$: shares of realised and of predicted “up”
- ▶ $\hat{P}^*$: share correct expected if predictions were independent of outcomes; $\widehat{SE}$: standard error of $\hat{P} - \hat{P}^*$ under H_0 (▶ Appendix)
- ▶ H_0 : independence; $\xrightarrow{d} N(0, 1)$: approximately standard Normal in large samples; $PT > 1.645$: skill at 5% (one-sided); $PT < 0$: fewer hits than chance
- With overlapping labels: slope t in $y_t = a + b x_t + u_t$, x_t the predicted and y_t the realised direction, Newey–West errors (Newey & West (1987)); $b > 0$: a predicted “up” is followed by more “up”
- **AUC**: the DeLong standard error (SE) assumes independent days; a circular block bootstrap resamples blocks of $L = 20$ consecutive days (Künsch (1989)) and keeps the serial dependence

Are the Differences Significant? Pesaran–Timmermann and AUC Intervals (2/2)

Pooled out-of-fold	Logit	RF	GBM	MLP
$\hat{P} / \hat{P}^*$	0.576 / 0.578	0.565 / 0.570	0.543 / 0.557	0.564 / 0.565
$PT / HAC\ t\ (\text{lag } 4)$	-1.28 / -1.00	-1.61 / -1.16	-3.07 / -2.31	-0.36 / -0.26
AUC	0.479	0.495	0.490	0.512
block-bootstrap 95% CI	[0.455; 0.504]	[0.472; 0.520]	[0.468; 0.514]	[0.489; 0.537]

- HAC t : the slope t -statistic with heteroskedasticity and autocorrelation consistent (HAC) errors, 4 lags
- No significant positive directional skill; every AUC interval contains 0.5; GBM is significantly worse than independence
- Consistent with weak-form efficiency ([Chapter 2](#)); look at cross-sections ([Gu, Kelly & Xiu \(2020\)](#)), risk targets, crypto delisting ([Będowska-Sójka, Wójcik & Pele \(2026\)](#)), text ([Chapter 15](#))

Comparing Forecasts: Diebold–Mariano, Clark–West, Giacomini–White (1/2)

- ▣ Two probability forecasts of $y_t \in \{0, 1\}$: model 1 gives $\hat{p}_{1,t}$, the benchmark (model 0, e.g. a constant) gives $\hat{p}_{0,t}$
- ▣ Loss of a forecast $\hat{p}_t$, lower is better: log-loss $L = -[y_t \log \hat{p}_t + (1 - y_t) \log(1 - \hat{p}_t)]$ or Brier loss $(y_t - \hat{p}_t)^2$
- ▣ Loss differential $d_t = L(y_t, \hat{p}_{1,t}) - L(y_t, \hat{p}_{0,t})$; $d_t > 0$: model 1 worse on day t
- ▣ **Diebold–Mariano** (DM, Diebold & Mariano (1995)): is the mean loss differential zero?

$$DM = \frac{\bar{d}}{\sqrt{\hat{\Omega}_{NW}/n}} \xrightarrow{d} N(0, 1) \quad \text{under } H_0 : \mathbb{E}[d_t] = 0$$

- ▶ $\bar{d}$: mean of d_t over the n test days; $\hat{\Omega}_{NW}$: heteroskedasticity and autocorrelation consistent (HAC) long-run variance of d_t , by Newey–West
- ▶ $DM > 1.96$: model 1 significantly worse at 5%; $DM < -1.96$: significantly better

Comparing Forecasts: Diebold–Mariano, Clark–West, Giacomini–White (2/2)

- ▣ **Nested models** (the logit with 14 features nests the constant, $\beta = 0$): under H_0 the large model estimates zeros with noise, $\mathbb{E}[d_t] > 0$, and DM rarely favours it

- ▣ **Clark–West** (CW, [Clark & West \(2007\)](#)): one-sided HAC t -test of $\mathbb{E}[f_t] = 0$ against $\mathbb{E}[f_t] > 0$

$$f_t = (y_t - \hat{p}_{0,t})^2 - [(y_t - \hat{p}_{1,t})^2 - (\hat{p}_{0,t} - \hat{p}_{1,t})^2]$$

- ▶ the Brier-loss gain of model 1, plus the term $(\hat{p}_{0,t} - \hat{p}_{1,t})^2$ that removes the noise cost of estimating zero coefficients
 - ▶ a large positive t : the features add information
 - ▶ derived for nested linear models estimated by OLS; for penalised-logit probabilities an approximation, not a theorem
- ▣ **Giacomini–White** (GW, [Giacomini & White \(2006\)](#)): tests the forecasting *method* as deployed, with a bounded estimation window; CW asks whether the population model adds information (statistic in the [Appendix](#))

Feature Importance: MDI, MDA and SHAP

Mean decrease impurity (MDI)

- Average Gini reduction from splits on a feature, over the trees
- In sample, trees only; favours continuous, high-cardinality features

Mean decrease accuracy (MDA)

- Drop in an out-of-sample score (here AUC) when a feature is randomly permuted
- Any model; can be negative: the feature hurts

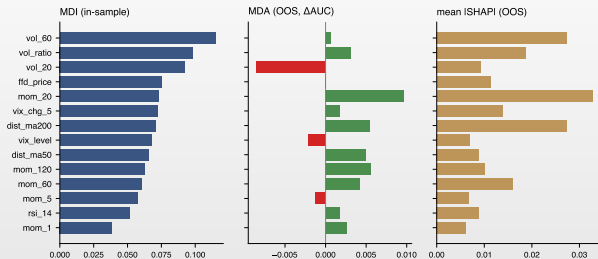
SHAP (Lundberg & Lee (2017))

- Shapley value ϕ_j : average marginal contribution of feature j over coalitions
- Additive: $\hat{f}(x) = \phi_0 + \sum_j \phi_j$, ϕ_0 the average prediction; mean $|\phi_j|$ = global importance

- MDI answers “what did the model use?”; MDA and SHAP answer “what helps on new data?”
- Correlated features share and dilute importance (substitution): cluster them first; coalition formula and the conditional vs. interventional choice in the [Appendix](#)

 Shapley Values

Feature Importance in the Case Study



- Fit on the first 70%, evaluated on the last 30% after a purge; **RF** gives MDI and MDA, **GBM** gives SHAP: the differences are not purely methodological
- MDI ranks volatility first; MDA and SHAP point to 20-day momentum and the distance from the 200-day moving average
- 20-day volatility: high MDI, **negative** MDA: a sign of overfitting, not a proof
- All importances are tiny ($\Delta AUC < 0.01$)
- Correlated features: permutation creates off-manifold points, so MDA and SHAP **extrapolate** (Hooker, Mentch & Zhou (2021))

 **MFM_ch13_feature_importance**

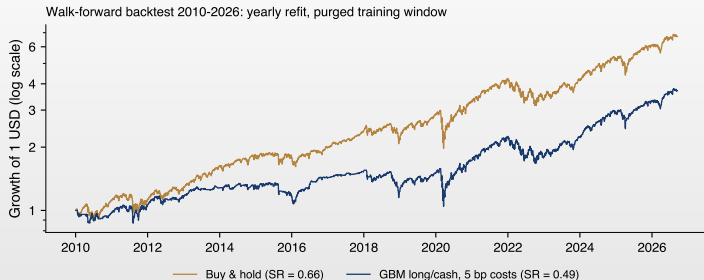
Walk-Forward Backtest: Strategy and Design

- Position $w_t \in \{0, 1\}$: long the S&P 500 if $\hat{p}_t > 0.5$, otherwise cash, decided with information up to t ; $\hat{p}_t$: the model's probability of "up"
- Strategy return: the index return when invested, minus a cost each time the position changes

$$r_{t+1}^S = w_t r_{t+1} - c |w_t - w_{t-1}|$$

- ▶ r_{t+1} : daily log return of the index; $r - c$ is the first-order approximation of $\log(e^r - c)$
- ▶ turnover $|w_t - w_{t-1}|$: 1 on the days the position changes; cost $c = 5$ bp (0.05%) per unit traded
- ▶ exposure $\bar{w} = \frac{1}{T} \sum_t w_t$: share of the T days spent invested
- Model: GBM refitted every January on all past data, purging labels that overlap the test year
- Out-of-sample period 2010–2026, one refit a year; benchmark buy-and-hold ($w_t = 1$: exposure 100%, no turnover)
- The only decision the backtest evaluates is the one a trader could have taken on day t

Walk-Forward Backtest: Results



- 2010–2026: strategy Sharpe **0.49** vs. buy-and-hold **0.66** (annualised); average exposure 77%; the model mostly learned “be long” and exited at the wrong times
- Difference of the two Sharpe ratios -0.17 (Lo SE ≈ 0.25 each, formula in the Sharpe-ratio inference slide): Ledoit–Wolf HAC SE 0.095, $p = 0.073$ (Ledoit & Wolf (2008)); block-bootstrap CI $[-0.36; 0.02]$: no evidence of skill, and no significant evidence that the strategy is worse
- With $d = 0.15$, chosen by the same rule on data up to 2009, SR = 0.43; without the FFD log-price 0.51

Walk-Forward: Do the Forecasts Beat a Constant?

- Walk-forward 2010–2026, $n = 4,198$ days, 60.2% “up”; benchmark: the constant forecast = training share of “up”
- Logit (nests the constant): log-loss 0.6822 vs. 0.6757
 - ▶ DM $t = 2.39$ ($p = 0.017$): the estimated logit is significantly *worse*
 - ▶ Clark–West $t = -0.21$ (one-sided $p = 0.58$): no evidence that the features carry information
 - ▶ Giacomini–White = 5.77 (χ^2_2 , $p = 0.056$): no significant difference at 5%; approximate, since the window expands (rolling-window result in the [Appendix](#))
- GBM: log-loss 0.6971; DM $t = 5.09$, GW = 25.8 ($p < 0.001$): worse than a constant; Pesaran–Timmermann -0.74
- An honest negative result is a result: no model beats “always up” or a constant forecast

Selection Bias and the False Strategy Theorem

- A researcher tries N strategies or configurations and reports the best; even if all have zero true Sharpe ratio, $\max_n \widehat{SR}_n$ grows with N : **selection bias**
- Harvey, Liu & Zhu (2016): given hundreds of published factors, a new factor needs $t > 3$, not 2

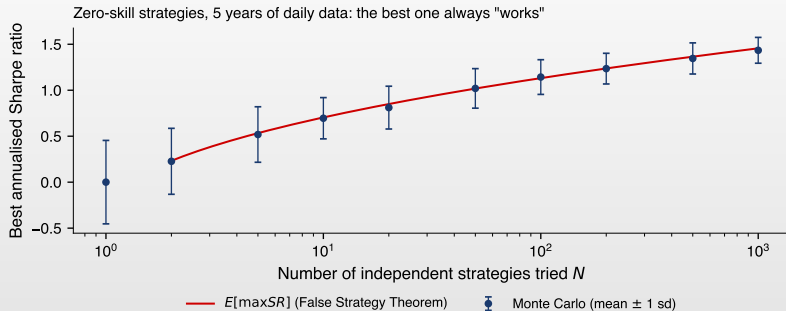
Theorem 1 (False Strategy Theorem, Bailey & López de Prado (2014))

- If $\widehat{SR}_n$, $n = 1, \dots, N$, are independent and approximately $N(0, \text{Var}[\widehat{SR}])$ (no skill), the **expected maximum Sharpe ratio** is

$$SR_0 \equiv \mathbb{E} \left[\max_n \widehat{SR}_n \right] \approx \sqrt{\text{Var}[\widehat{SR}]} \left[(1 - \gamma) \Phi^{-1} \left(1 - \frac{1}{N} \right) + \gamma \Phi^{-1} \left(1 - \frac{1}{Ne} \right) \right]$$

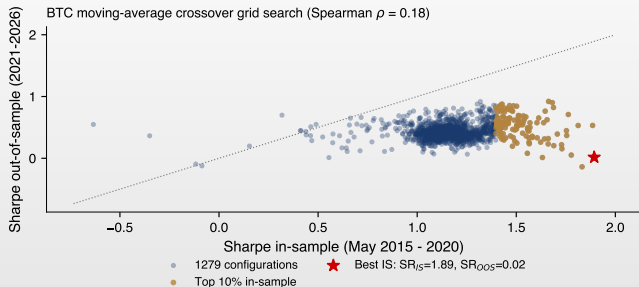
- Notation
 - ▶ $\widehat{SR}_n$: estimated Sharpe ratio of strategy n ; $\text{Var}[\widehat{SR}]$: sampling variance of one estimated SR; i.i.d. returns: $\approx 1/T$ per period, T the number of return periods
 - ▶ Φ : standard Normal distribution function; Φ^{-1} : its quantile function; $e \approx 2.718$; $\gamma \approx 0.5772$: Euler–Mascheroni constant
- Reading: SR_0 is the Sharpe ratio the luckiest of N worthless strategies is expected to show; it rises with N and with the sampling noise $\sqrt{\text{Var}[\widehat{SR}]}$

The Best of N Zero-Skill Strategies



- Strategies with i.i.d. zero-mean returns, 5 years of daily data (annualised $\text{Var}[\widehat{SR}] \approx 1/5$)
- Trying $N = 1000$ variants yields a best Sharpe ratio of ≈ 1.45 — **with no skill at all**
- The Monte Carlo simulation agrees with the theorem

In-Sample vs. Out-of-Sample: Bitcoin Grid Search



- Moving-average (MA) crossover on Bitcoin: long when the fast MA of the log price exceeds the slow MA by more than a band; 1,279 configurations (fast, slow, band); in sample 15 May 2015 – 2020, out of sample 2021–2026
- Selected rule (14/20 days): SR_{IS} **1.89** → SR_{OOS} **0.02** (annualised with $\sqrt{365}$); Spearman rank correlation IS–OOS $\rho = 0.18$
- Median OOS SR 0.55 for the top 10% in sample vs. 0.42 for all: the *family* of trend rules carries information, the *selected* optimum is noise

Sharpe Ratio Inference: Standard Error and PSR (1/2)

- ▣ Per-period Sharpe ratio (SR) of daily log returns, risk-free rate 0: $\widehat{SR} = \bar{r}/\hat{\sigma}$, the mean return $\bar{r}$ over its standard deviation $\hat{\sigma}$
- ▣ Shape of the return distribution, with μ , σ the mean and SD of r
 - ▶ skewness $\gamma_3 = \mathbb{E}[(r - \mu)^3]/\sigma^3$: < 0 when large losses are more frequent than large gains
 - ▶ ordinary (not excess) kurtosis $\gamma_4 = \mathbb{E}[(r - \mu)^4]/\sigma^4$: 3 for the Normal distribution, larger for fat tails
- ▣ Sampling variance of $\widehat{SR}$, i.i.d. returns, delta method (Lo (2002); Opdyke (2007)):

$$\text{Var}(\widehat{SR}) \approx \frac{1}{T} \left(1 - \gamma_3 SR + \frac{\gamma_4 - 1}{4} SR^2 \right)$$

- ▶ T : number of returns; SR : true per-period Sharpe ratio
- ▶ negative skewness and fat tails widen the uncertainty; Normal distribution ($\gamma_3 = 0$, $\gamma_4 = 3$): $(1 + \frac{1}{2}SR^2)/T$
- ▣ Annualising with $\sqrt{A}$, A return periods per year (252 trading days; 365 for Bitcoin), needs zero autocorrelation; otherwise $SR_A = \eta(A) SR$ (Lo (2002))
 - ▶ $\eta(A) = A/\sqrt{A + 2 \sum_{k=1}^{A-1} (A-k)\rho_k}$, ρ_k : autocorrelation of returns at lag k ; $\rho_k > 0$ makes $\eta(A) < \sqrt{A}$

Sharpe Ratio Inference: Standard Error and PSR (2/2)

Definition 5 (Probabilistic Sharpe Ratio (PSR), Bailey & López de Prado (2012))

▣ Probability that the true SR exceeds a benchmark SR^* , the variance above in the denominator:

$$\widehat{PSR}(SR^*) = \Phi \left(\frac{(\widehat{SR} - SR^*)\sqrt{T-1}}{\sqrt{1 - \hat{\gamma}_3 \widehat{SR} + \frac{\hat{\gamma}_4 - 1}{4} \widehat{SR}^2}} \right)$$

- ▣ SR^* : the benchmark, e.g. 0 or the expected maximum SR_0 ; $\hat{\gamma}_3$, $\hat{\gamma}_4$: sample skewness and kurtosis
- ▣ The argument of Φ is a z-statistic: the distance $\widehat{SR} - SR^*$ in standard errors
- ▣ Reading: $\widehat{PSR} > 0.95$: the true SR exceeds SR^* with 95% confidence
- ▣ Two strategies on the same days: test $SR_a - SR_b$ with the HAC delta method or a studentised block bootstrap (Ledoit & Wolf (2008)), not with two separate intervals

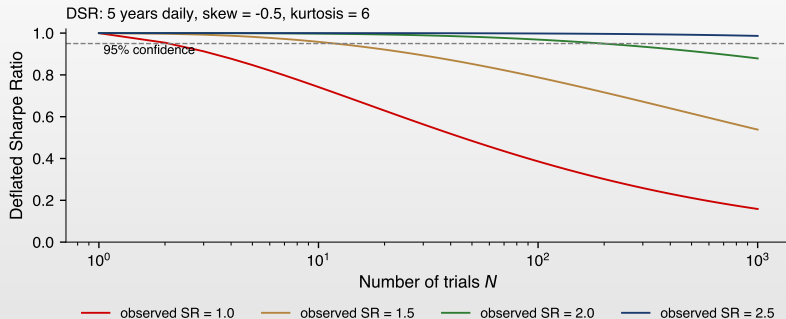
Deflated Sharpe Ratio: the Best Bitcoin Rule

- **Deflated Sharpe Ratio** (DSR, Bailey & López de Prado (2014)): $DSR = \widehat{PSR}(SR_0)$, the PSR with the benchmark set to SR_0 , the expected maximum of the False Strategy Theorem computed with $\text{Var}[\widehat{SR}_n]$ of the trials
 - ▶ corrects for selection (N trials), non-Normality and sample length; reject unless $DSR > 0.95$, and report N
- Worked example, best in-sample rule: $T = 2,058$ days, $\widehat{SR} = 0.0991$ a day (1.89 annualised), $\hat{\gamma}_3 = 0.12$, $\hat{\gamma}_4 = 12.33$
 - ▶ denominator $\sqrt{1 - 0.12 \cdot 0.0991 + \frac{12.33-1}{4} \cdot 0.0991^2} = 1.008$; $N = 1,279$ trials
 - ▶ $z = (\widehat{SR} - SR_0)\sqrt{T-1}/1.008$, the argument of Φ ; $DSR = \Phi(z)$; $SD = \sqrt{\text{Var}[\widehat{SR}_n]}$

Var $[\widehat{SR}_n]$ from	SD	SR_0 (per day)	z / DSR
dispersion of the 1,279 trial SRs	0.0106	0.035	2.88 / 0.998
independent trials, $1/T$	0.0220	0.073	1.16 / 0.877

- The trials are highly correlated, so their dispersion understates the search: the verdict depends on this choice, and the out-of-sample SR of 0.02 sides with the conservative value

Deflated Sharpe Ratio in Practice



- 5 years of daily returns, skewness -0.5 , kurtosis 6
- An annualised SR of 1.0 is significant only if (almost) no other strategy was tried: it fails after 2–3 trials
- $SR = 1.5$ survives ≈ 10 trials, $SR = 2.0 \approx 200$; only $SR = 2.5$ survives 1000

Data-Snooping Tests on the Bitcoin Grid (1/2)

- ▣ K rules; $r_{k,t}$: return of rule k on day t ; $r_{0,t}$: benchmark return; differential $d_{k,t} = r_{k,t} - r_{0,t}$; $\bar{d}_k$: its mean over the n days
- ▣ $H_0 : \max_k \mathbb{E}[d_{k,t}] \leq 0$: no rule beats the benchmark
- ▣ Null distribution from a **stationary bootstrap** (Politis & Romano (1994)): resample blocks of whole days, of random length, the same days for all rules
 - ▶ keeps the correlation between rules: no “effective” N needed
- ▣ Reality Check (RC, White (2000)): statistic $\max_k \sqrt{n} \bar{d}_k$, the scaled mean of the best rule; p -value: share of bootstrap maxima above it
- ▣ Superior Predictive Ability (SPA, Hansen (2005)): studentised, $\max_k \sqrt{n} \bar{d}_k / \hat{\omega}_k$ with $\hat{\omega}_k$ the bootstrap SD of $\sqrt{n} \bar{d}_k$; clearly bad rules not recentred
- ▣ Romano–Wolf StepM (Romano & Wolf (2005)): names *which* rules win, with family-wise error rate (FWER) control, the probability of at least one false rejection

Data-Snooping Tests on the Bitcoin Grid (2/2)

- ▣ Setup: 1,279 rules, in-sample $n = 2,058$ days, mean block 20 days, 2,000 replications
- ▣ Implementation details in the [Appendix](#)

Benchmark	largest studentised mean	RC p	SPA p	StepM rules kept
cash ($r_0 = 0$)	—	0.004	0.002	314
buy-and-hold	0.88 (the 14/20 rule)	0.74	0.70	0

- ▣ Against cash, some trend rule earns a positive mean: Bitcoin's own trend; no rule beats holding Bitcoin, even in sample, consistent with the OOS SR of 0.02
- ▣ DSR asks “above the luck of N trials?” and needs N and $\text{Var}[\widehat{SR}]$ (0.998 or 0.877); SPA takes both from the data

Probability of Backtest Overfitting (PBO)

Combinatorially symmetric cross-validation (CSCV): Bailey, Borwein, López de Prado & Zhu (2017)

1. Matrix of returns: T periods $\times$ N configurations; split the rows into S contiguous blocks
 2. For each of the $\binom{S}{S/2}$ choices, half of the blocks are in sample (IS), the rest out of sample (OOS)
 3. Pick the best IS configuration n^* ; rank all N configurations OOS in ascending order and set $\bar{\omega} = \text{rank}(n^*)/(N + 1) \in (0, 1)$
 4. Logit $\lambda = \log \frac{\bar{\omega}}{1-\bar{\omega}}$; $PBO = \Pr[\lambda \leq 0]$: the share of splits in which the IS winner ranks at or below the OOS median
- $\binom{S}{S/2}$: number of ways to choose $S/2$ of the S blocks; $\bar{\omega}$: relative OOS rank of the IS winner, near 1 if it stays the best; $\lambda > 0$: above the OOS median
 - $PBO \approx 0$: the IS winner reliably stays above the OOS median; $PBO \approx 0.5$ or more: selecting the IS winner is no better than picking at random
 - PBO evaluates the **research procedure**, not one strategy, and is not the probability of losing money; like DSR it needs **every** trial logged (Seminar B11)

Selection Is Not Inference: Post-Double Selection (1/2)

- The LASSO path answers “which feature enters first”, not “which feature matters”; OLS t -statistics after LASSO ignore the selection step
- A control correlated with the regressor of interest d_t but only weakly with y_t gets dropped: its effect moves into the coefficient of d_t (omitted-variable bias)
- Partially linear model: the effect of d_t is linear, the controls enter through unknown functions

$$y_t = \theta d_t + g(x_t) + \varepsilon_t, \quad d_t = m(x_t) + v_t$$

- ▶ y_t : outcome; d_t : regressor of interest; θ : its effect, the parameter to estimate
- ▶ x_t : the p controls; g : their effect on y_t ; m : how they predict d_t ; ε_t , v_t : errors
- ▶ approximately sparse: a few controls explain most of g and m

Selection Is Not Inference: Post-Double Selection (2/2)

- Post-double selection (Belloni, Chernozhukov & Hansen (2014)):
 1. LASSO of y_t on $x_t \Rightarrow$ selected set $\hat{S}_y$
 2. LASSO of d_t on $x_t \Rightarrow$ selected set $\hat{S}_d$
 3. OLS of y_t on d_t and $x_{\hat{S}_y \cup \hat{S}_d}$, robust standard error
- $\hat{S}_y, \hat{S}_d$: indices of the controls kept by each LASSO; $x_{\hat{S}_y \cup \hat{S}_d}$: the controls kept by either
- A control is missed only if it matters for *neither* equation, so the omitted bias is second order
- Valid under independent sampling and sparsity, with HAC errors an approximation for time series (penalty and conditions in the [Appendix](#))

Double/Debiased ML: Orthogonal Scores and Cross-Fitting

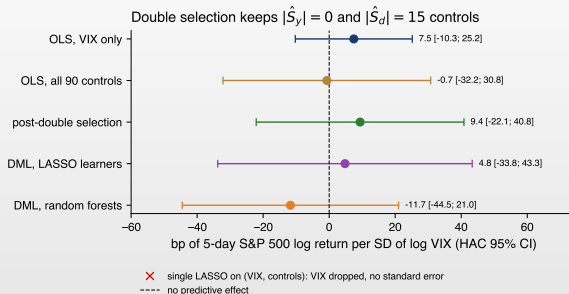
- Chernozhukov et al. (2018): residualise both variables with *any* learner (LASSO, RF, boosting), then regress residual on residual

$$\tilde{y}_t = y_t - \hat{\ell}(x_t), \quad \tilde{v}_t = d_t - \hat{m}(x_t), \quad \hat{\theta} = \frac{\sum_t \tilde{v}_t \tilde{y}_t}{\sum_t \tilde{v}_t^2}$$

- ▶ $\hat{\ell}(x) = \hat{\mathbb{E}}[y \mid x]$: ML prediction of y_t from the controls; $\hat{m}(x)$: ML prediction of d_t
- ▶ $\tilde{y}_t, \tilde{v}_t$: the parts of y_t and d_t the controls do not explain; $\hat{\theta}$: OLS slope of $\tilde{y}_t$ on $\tilde{v}_t$

- Neyman-orthogonal score $\psi = (\tilde{y} - \theta \tilde{v}) \tilde{v}$, whose mean is zero at the true θ
 - ▶ first-order errors in $\hat{\ell}$ and $\hat{m}$ do not bias $\hat{\theta}$ (rate conditions in the [Appendix](#))
- Cross-fitting: fit $\hat{\ell}, \hat{m}$ on $K - 1$ folds, residualise the held-out fold, repeat; removes the own-observation overfitting bias
- The published theory assumes independent sampling; with overlapping 5-day labels we use contiguous folds purged by 5 days around the test fold and HAC standard errors, a data limitation stated as such, not a new estimator

Worked Example: Does the VIX Predict S&P 500 Returns?



- y_t : 5-day forward log return (bp); d_t : log VIX, standardised (1 SD = 0.356); 90 controls: 12 technical features, squares, pairwise products; $n = 6,269$; HAC errors, 4 lags
- The single LASSO drops the VIX and every control: “no effect”, with no standard error
- Double selection keeps $|\hat{S}_y| = 0$ and $|\hat{S}_d| = 15$ controls ($|\cdot|$: number of elements); 95% CI for θ $[-22.1; 40.8]$ bp per week per SD
- Honest intervals contain zero and large effects of both signs: the data cannot decide

The Virtue of Complexity, and a Critical Reading

- Kelly, Malamud & Zhou (2024): ridge regressions on P random Fourier features of a few predictors, rolling windows as short as $T = 12$ months, P in the thousands
 - ▶ P : number of features (the paper's notation); random Fourier features: sin and cos of random linear combinations of the predictors; T : training window, in months
 - ▶ theory: with $P > T$ the minimum-norm solution (the exact fit with the smallest $\|\beta\|_2$) shrinks implicitly ("double descent"); R_{OOS}^2 may be negative while the Sharpe ratio of market timing **rises** with complexity; complexity is not automatically overfitting
- Nagel (2025): for $P \gg T$ the forecast is a similarity-weighted average of the T training returns; in short windows similarity is mostly recency, so the method becomes a **volatility-timed momentum** strategy, which loses on artificial data with reversals
- Avramov, Cheng & Metzker (2023): deep-learning signal profits come from hard-to-arbitrage stocks; excluding microcaps, distressed firms or high-volatility episodes attenuates them, and costs erode them further
- Chen, Pelger & Zhu (2024): impose economic structure instead; the no-arbitrage moment conditions become the loss of a deep network (next slides)
- Survey of the field: Kelly & Xiu (2023)

Case Study: Deep Learning in Asset Pricing, Question and Method (1/2)

- ▣ [Chen, Pelger & Zhu \(2024\)](#): estimate the stochastic discount factor (SDF) M_{t+1} that prices **all individual US stocks**, conditioning on 46 firm characteristics and on the history of 178 macroeconomic series
- ▣ No arbitrage: the SDF prices every excess return (Section I.A, eq. 1)

$$E_t[M_{t+1}R_{t+1,i}^e] = 0, \quad M_{t+1} = 1 - \omega_t^\top R_{t+1}^e$$

- ▶ $R_{t+1,i}^e$: excess return of stock i over the Treasury bill from month t to $t+1$; R_{t+1}^e : the vector of all stocks
- ▶ E_t : expectation given the information at t ; ω_t : portfolio weights, a function of characteristics and macro states
- ▣ $F_{t+1} = \omega_t^\top R_{t+1}^e$, the SDF portfolio, is the conditional tangency portfolio (highest Sharpe ratio), and

$$E_t[R_{t+1,i}^e] = \beta_{t,i} E_t[F_{t+1}]$$

- ▶ $\beta_{t,i}$: loading of stock i on F ; a stock's expected excess return is its loading times the premium of F

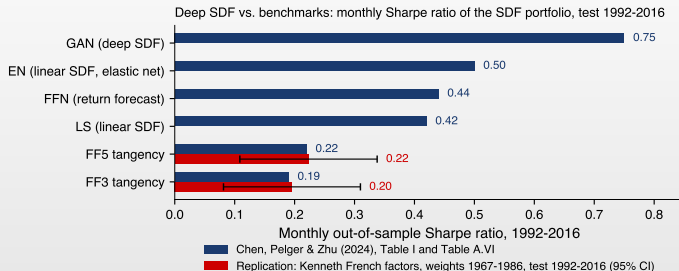
Case Study: Deep Learning in Asset Pricing, Question and Method (2/2)

- Why: forecasting (Gu, Kelly & Xiu (2020), the benchmark feedforward network, FFN) fits a second moment where signal-to-noise is low; the no-arbitrage condition targets a first moment, and averaging over time removes much of the noise
- Method: adversarial generalized method of moments (GMM, eq. 3, after Hansen & Jagannathan (1997))

$$\min_{\omega} \max_g \frac{1}{N} \sum_{j=1}^N \left(E \left[M_{t+1} R_{t+1,j}^e g(I_t, I_{t,j}) \right] \right)^2$$

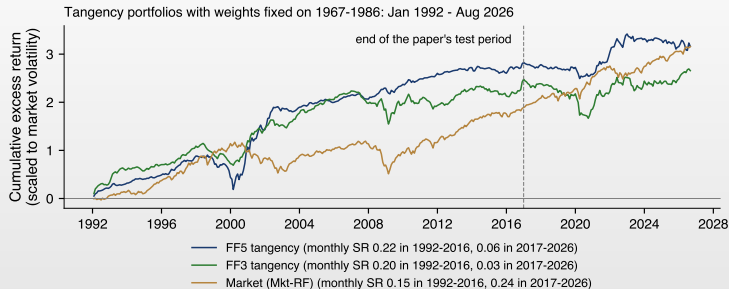
- ▶ I_t : macroeconomic information; $I_{t,j}$: characteristics of stock j ; N : number of stocks
- ▶ g : instrument function chosen by an adversary to make the pricing errors as large as possible; ω minimises this worst case
- Networks: an FFN for ω , a long short-term memory network (LSTM) that turns the macro history into 4 states, an adversary g that builds the hardest test assets; the full model is labelled GAN
- Chronological split: training 1967–1986, validation 1987–1991, test 1992–2016; configurations ranked by the validation Sharpe ratio (data and tuning in the [Appendix](#))

Deep Learning in Asset Pricing: Sharpe Ratios, Paper and Replication



- Blue: the paper, monthly out-of-sample SR 1992–2016: GAN 0.75, elastic net linear SDF (EN) 0.50, FFN 0.44, unregularised linear SDF (LS) 0.42 (Table I); FF5 0.22, FF3 0.19 (Table A.VI); the regularised no-arbitrage model already beats the forecasting network
- Red: FF3 and FF5 tangency portfolios $w \propto \hat{\Sigma}^{-1} \hat{\mu}$ on the [Kenneth French](#) factors, weights fixed on 1967–1986: FF5 0.22 [0.11; 0.34], FF3 0.20 [0.08; 0.31] (SE 0.058, [Lo \(2002\)](#)): both published values lie inside the intervals
 - $\hat{\mu}$, $\hat{\Sigma}$: sample mean vector and covariance matrix of the factor returns; $\propto$: proportional to

Deep Learning in Asset Pricing: the Benchmarks after 2016



- Same fixed weights, January 1992 – August 2026, each series scaled to the volatility of the market excess return on 1992–2016; monthly SR 2017–2026 ($T = 116$, $SE \approx 0.09$): FF5 0.06, FF3 0.03, market 0.24; on 1992–2016: 0.22, 0.20, 0.15
- Replicable with public data: the Fama–French benchmarks; not replicable: the stock-level GAN, LS and EN (the CRSP panel is not in the course data); the post-2016 fall is a warning, not yet a test, and the same question applies to a GAN frozen in 1991 (details in the [Appendix](#))

An Open Question: Discovery or Data Mining at Scale?

- ▣ [Chen, Lopez-Lira & Zimmermann \(2022\)](#): mining 29,000 accounting ratios for $t > 2$ gives cross-sectional predictability similar to peer-reviewed predictors; for both, about half remains after the original samples
- ▣ [Novy-Marx & Velikov \(2025\)](#): an LLM (large language model) pipeline mines over 30,000 signals, keeps 96 that pass a fixed protocol and writes hundreds of complete papers, with invented motivations and some imagined citations: HARKing (hypothesising after the results are known) at scale
- ▣ Open: when a machine proposes features, model and story
 - ▶ what is the number of trials?
 - ▶ what separates a discovery from the best of many draws?
- ▣ The discovery loop: literature → competing hypotheses → data and code → robustness fixed in advance → referee-style critique → report every test (reusable prompts in the [Appendix](#))
- ▣ What the human checks: every reference resolves (DOI, arXiv); every trial is logged, so M in the correction counts them all; correlated tests get a joint null; transformations live inside the training fold

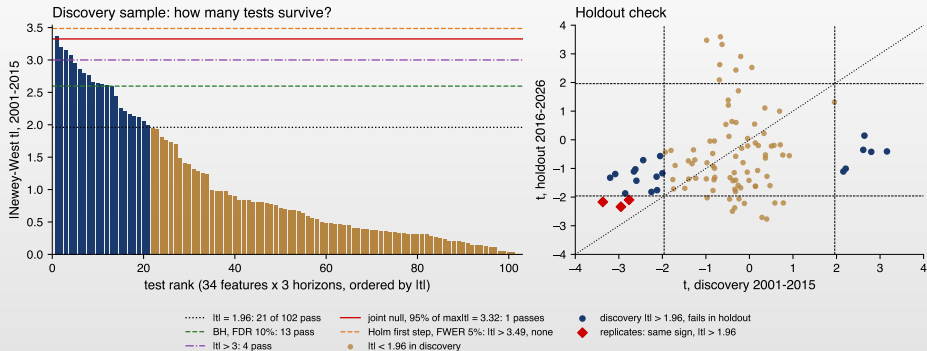
Mini-Case: 102 Tests, One Old Survivor (1/2)

- A typical assistant proposal: “add more indicators: momentum, moving-average distances, volatility ratios, RSI and VIX changes over several windows”
- 34 features $\times$ 3 horizons (1, 5, 20 days) = 102 predictive regressions on 2001–2015, Newey–West t
 - ▶ regression j : $r_{t,t+h} = \alpha_j + \beta_j x_{j,t} + e_t$, with $x_{j,t}$ feature j and $r_{t,t+h}$ the next h -day return; $H_{0,j} : \beta_j = 0$
- Corrections for $M = 102$ tests, $p_{(1)} \leq \dots \leq p_{(M)}$ the ordered p -values
 - ▶ Holm: reject while $p_{(j)} \leq 0.05/(M - j + 1)$; controls the FWER, the probability of any false rejection
 - ▶ Benjamini–Hochberg (BH, [Benjamini & Hochberg \(1995\)](#)): reject the j smallest, for the largest j with $p_{(j)} \leq j q/M$; controls the false discovery rate (FDR) at q , the expected share of false rejections among all rejections
 - ▶ joint null: the 95% quantile of $\max_j |t_j|$ when the target is shifted circularly, which keeps the correlation between the tests

Mini-Case: 102 Tests, One Old Survivor (2/2)

- ▣ $|t| > 1.96$: **21** of 102, against 5.1 expected under the global null at 5%
- ▣ Holm (FWER 5%): **0**; BH (FDR 10%): **13**; $|t| > 3$ (Harvey, Liu & Zhu (2016)): **4**
- ▣ Joint null (500 circular shifts of the target): 95% quantile of $\max_j |t_j| = 3.32$; one survivor, the 1-day return on the next day's return, $t = -3.37$: short-term reversal, a known effect
- ▣ Holdout 2016–2026: 3 of the 21 keep their sign with $|t| > 1.96$, all reversal variants; the VIX-based “finds” fail
- ▣ Holm and BH control different error rates at different levels; BH needs independence or positive dependence of the true-null p-values, which correlated predictors do not ensure

Mini-Case: Thresholds and the Holdout



- Left: the 102 discovery $|t|$ in decreasing order; each threshold answers a different question, and only the joint null accounts for the correlation between the tests
- Right: discovery vs. holdout t ; the three replications (red) are all short-term reversal; the significant VIX features collapse towards zero

Your Turn: a Project Seed

▣ Questions

- ▶ can an AI-assisted feature search find a predictor of index returns (S&P 500, BET, Bitcoin)?
- ▶ does it survive a pre-registered correction and a holdout?
- ▶ does it go beyond short-term reversal?

▣ Replicate first: this mini-case (21 of 102, one survivor at the joint null)

▣ Deliverable: a feature list fixed before the data are touched; all tests with Holm, BH and the $\max\text{-}|t|$ null; a holdout; DSR and PBO for any strategy built on the survivors

▣ Log: each prompt that shaped a decision in AI_USE.md; each invented reference, leak or undercounted trial in AI_ERRORS.md, with the test that caught it ([templates](#))

Key Takeaways and a Decision Checklist

Three empirical lessons

- An ADF rejection is not stationarity: truncated FFD of the S&P 500 log price stays $I(1)$; estimate memory with local Whittle
- Shuffled K -fold reports 69% on pure noise; on the S&P 500 no model beats “always up” (PT, DM, GW)
- The best of many backtests must be deflated or bootstrapped: Bitcoin grid 1.89 in sample, 0.02 out of sample

Next

- Chapter 14: deep learning and time-series foundation models; Chapter 15: large language models and sentiment

Before any claim

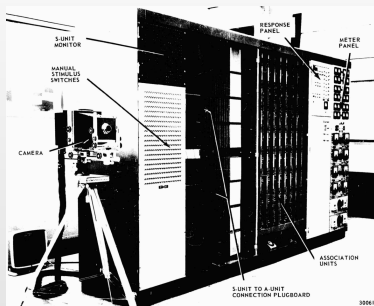
- Point-in-time data; transformations fitted on training folds only
- Volatility-scaled labels with recorded spans
- Purged K -fold or CPCV with embargo; walk-forward for the final test
- A naive baseline; PT, DM / CW / GW and block-bootstrap intervals
- Costs included; every trial logged; DSR, SPA and PBO
- “Selected” or “important” is not “significant”: double selection or DML for inference

Appendix: Further Study on Quantinar

Advanced courses on quantinar.com

- ☐  Random Forests
- ☐  Shapley Values
- ☐  Shapley Curves: A Smoothing Perspective
- ☐  Machine Learning in Financial Risk
- ☐  Introduction to Reinforcement Learning
- ☐  Generative Adversarial Networks
- ☐  MSCA Digital: The Need for Explainable AI
- ☐  Business Analytics and Data Science
- ☐  Model Selection Criteria
- ☐  PAC Learning

Appendix: From the Perceptron to Deep Blue



Mark I Perceptron, operator's manual (1960)

Figure: J.C. Hay, A.E. Murray (1960); public domain; Wikimedia Commons



IBM Deep Blue, Computer History Museum

Photo: Anton Chiang (2011); CC BY 2.0; Wikimedia Commons

- **Perceptron:** a learning rule for a single artificial neuron [Rosenblatt \(1958\)](#); the Mark I was hardware built to recognise simple images
- 1997: IBM's Deep Blue beats world chess champion Garry Kasparov [Campbell, Hoane & Hsu \(2002\)](#)
- Deep Blue searched with hand-built rules; modern ML (this chapter) *learns* the rules from data

Appendix: the In-Sample R^2 of a Predictor without Skill

- ▣ A monthly R^2 of 0.5% is economically large, but sampling noise is of the same order
- ▣ In-sample OLS R^2 of one predictor with no skill (intercept, independent errors, $n \approx 300$ months): $nR^2 \approx \chi_1^2$
 - ▶ n : number of months; χ_1^2 : chi-squared distribution with 1 degree of freedom, mean 1 and variance 2
 - ▶ so R^2 has mean $1/n \approx 0.3\%$ and standard deviation $\sqrt{2}/n \approx 0.5\%$
- ▣ R_{OOS}^2 can be negative, has no such null distribution and needs its own test (Clark–West)

Appendix: Worked Example: How a Forest Averages Its Trees

- ▣ Random forest of the case study (400 trees, minimum leaf size 100), trained on the first four purged folds
- ▣ Prediction for the last observation (11 September 2026): $P(\text{up in 5 days})$ from each tree
 - ▶ individual trees range from 0.31 to 0.88 (standard deviation 0.11)
 - ▶ 81.5% of the trees vote “up” ($p > 0.5$)
 - ▶ forest probability = average of the 400 trees = 0.608
- ▣ Single trees disagree strongly; the average is far more stable
- ▣ Stability is not accuracy: the forest still does not beat “always up” (case study)

Appendix: Sampling Financial Data: Bars

Time and tick bars

- ▣ A **bar** = one row (open, high, low, close, volume) summarising a sampling interval
- ▣ Time bars: every fixed interval (1 minute, 1 day); oversample quiet periods; fat-tailed returns
- ▣ Tick bars: one bar every N transactions

Volume, dollar and information bars

- ▣ One bar every V shares or every D dollars traded: sampling by **information arrival**
- ▣ Returns closer to i.i.d. draws from the Normal distribution: [Clark \(1973\)](#); [Ané & Geman \(2000\)](#)
- ▣ Imbalance / runs bars: sample when order flow becomes one-sided

- ▣ This chapter uses daily time bars; tick data (crypto exchanges) in [Chapters 10 and 16](#)

Appendix: Fractional Weights and Their Asymptotics

Definition 6 (Fractional difference operator)

- With the backshift operator $BX_t = X_{t-1}$ and a real order d :

$$(1 - B)^d = \sum_{k=0}^{\infty} \binom{d}{k} (-B)^k, \quad \tilde{X}_t = \sum_{k=0}^{\infty} w_k X_{t-k}$$

- Weights: $w_0 = 1$, $w_k = -w_{k-1} \frac{d-k+1}{k} = \frac{\Gamma(k-d)}{\Gamma(-d)\Gamma(k+1)} \sim \frac{k^{-(1+d)}}{\Gamma(-d)}$ as $k \rightarrow \infty$
- $\binom{d}{k} = d(d-1)\cdots(d-k+1)/k!$: the binomial coefficient, defined for any real d ; Γ : the gamma function, $\Gamma(k+1) = k!$
- $\sim$: the ratio of the two sides tends to 1; the weights decay like $k^{-(1+d)}$, more slowly for smaller d
- $d = 1$: $w = (1, -1, 0, 0, \dots)$ — ordinary returns
- $0 < d < 1$: weights decay hyperbolically $\Rightarrow$ long memory is kept
- Infinite sum: $\sum_{k=0}^{\infty} w_k = (1-1)^d = 0$, so the level of X_t cancels only in the limit

Appendix: Local Whittle and Exact Local Whittle (1/2)

- ▣ Periodogram: the sample estimate of the spectral density at the Fourier frequency $\lambda_j = 2\pi j/n$

$$I(\lambda_j) = \frac{1}{2\pi n} \left| \sum_{t=1}^n X_t e^{-i\lambda_j t} \right|^2$$

- ▣ Local Whittle ([Robinson \(1995\)](#)): choose δ to fit $f(\lambda) \approx G\lambda^{-2\delta}$ on the first m frequencies

$$\hat{\delta} = \arg \min_{\delta} \log \left(\frac{1}{m} \sum_{j=1}^m \lambda_j^{2\delta} I(\lambda_j) \right) - \frac{2\delta}{m} \sum_{j=1}^m \log \lambda_j, \quad \sqrt{m}(\hat{\delta} - \delta) \xrightarrow{d} N(0, \frac{1}{4})$$

- ▶ n : sample size; m : number of frequencies used; i : the imaginary unit; G has been profiled out
- ▶ $\xrightarrow{d}$: convergence in distribution; hence the standard error $1/(2\sqrt{m})$

Appendix: Local Whittle and Exact Local Whittle (2/2)

- Exact local Whittle (ELW, [Shimotsu & Phillips \(2005\)](#); unknown mean: [Shimotsu \(2010\)](#)): periodogram of $(1 - B)^\delta X_t$, valid also for non-stationary $\delta \geq \frac{1}{2}$
- Why the FFD series looks like $\hat{\delta} \approx 0.75$: above frequency $1/K$ the truncated filter acts as $(1 - B)^{0.25}$; only $n/(2\pi K) \approx 2$ of the $m = 294$ frequencies lie below $1/K$, where the unit root returns
 - ▶ K : the FFD window length (444 lags)
 - ▶ the estimate is the apparent memory over the band, not the integration order of the series

 MFM_ch13_ffd

Appendix: Worked Example: Purging with 10 Observations

- Observations $t = 0, \dots, 9$; label of t ends at $t + 2$ (capped at 9)
- 5 contiguous folds of 2; embargo = 1 observation

Test	Removed	Train
0, 1	2, 3	4–9
2, 3	0, 1, 4, 5	6–9
4, 5	2, 3, 6, 7	0, 1, 8, 9
6, 7	4, 5, 8, 9	0–3
8, 9	6, 7	0–5

- Test $\{4, 5\}$: labels of 2 and 3 end at 4 and 5 $\Rightarrow$ purged; 6 is the embargo; 7 starts when the last test label ends $\Rightarrow$ purged
- With overlapping labels, up to 4 of 10 observations leave the training set

Appendix: When Is K-Fold Valid for Time Series?

- Bergmeir, Hyndman & Koo (2018): for a **purely autoregressive** model $y_t = g(y_{t-1}, \dots, y_{t-p}) + \varepsilon_t$ with serially uncorrelated errors, standard K -fold CV is valid
 - ▶ g : any, possibly non-linear, function of the last p values; ε_t : error
 - ▶ the errors in the test fold are uncorrelated with the training errors, so the CV error is unbiased for the OOS error
- It fails when the **errors** are serially correlated
 - ▶ overlapping labels: h -day forward returns share $h - 1$ daily returns
 - ▶ persistent features that are not in the model's lag set (rolling means, levels such as the FFD log-price)
 - ▶ under-specified dynamics (too few lags)
- The leakage experiment isolates exactly this: 20-day labels and 20-day rolling features
- Check before choosing: the autocorrelation of the CV residuals; if it is significant, purge and embargo

Appendix: Walk-Forward and Combinatorial Purged CV

Walk-forward (WF)

- Train on $[0, T_k]$, test on $(T_k, T_{k+1}]$, roll forward; T_k : end of the k -th training window
- Mimics real-time deployment; no future data in training
- But a **single path**: high variance, easy to overfit by repeated trials

Combinatorial purged CV (CPCV)

- N contiguous groups; test on all $\binom{N}{k}$ combinations of k groups, with purging and embargo
- $\varphi = \frac{k}{N} \binom{N}{k}$: the number of complete backtest paths; e.g. $N = 6, k = 2$: 15 splits, 5 paths
- A **distribution** of Sharpe ratios instead of one number

- $\binom{N}{k}$: number of ways to choose k of N groups; each group is tested φ times, so the test predictions form φ full histories
- Rule of thumb: purged K-fold / CPCV for model selection; WF for the final, realistic evaluation

Appendix: Testing Directional Accuracy: Pesaran–Timmermann (1/2)

□ H_0 : the predicted sign x_t is independent of the realised sign y_t (Pesaran & Timmermann (1992))

□ $\hat{P}$ = share of correct signs; under H_0 the expected share is $\hat{P}^* = \hat{p}_y \hat{p}_x + (1 - \hat{p}_y)(1 - \hat{p}_x)$

▶ $\hat{p}_y, \hat{p}_x$: shares of realised and of predicted “up”; n : number of forecasts

▶ $\hat{p}_y \hat{p}_x$: chance that both are “up” if independent; $(1 - \hat{p}_y)(1 - \hat{p}_x)$: both “down”

$$PT = \frac{\hat{P} - \hat{P}^*}{\sqrt{\widehat{V}(\hat{P}) - \widehat{V}(\hat{P}^*)}} \xrightarrow{d} N(0, 1), \quad \widehat{V}(\hat{P}) = \frac{\hat{P}^*(1 - \hat{P}^*)}{n}$$

▶ $\widehat{V}(\hat{P}^*) = [(2\hat{p}_y - 1)^2 \hat{p}_x(1 - \hat{p}_x) + (2\hat{p}_x - 1)^2 \hat{p}_y(1 - \hat{p}_y)]/n + 4\hat{p}_y \hat{p}_x(1 - \hat{p}_y)(1 - \hat{p}_x)/n^2$

▶ $\widehat{V}(\cdot)$: estimated variance under H_0 ; $PT > 1.645$: directional skill at 5% (one-sided)

Appendix: Testing Directional Accuracy: Pesaran–Timmermann (2/2)

- Logit above: $\hat{p}_y = 0.581$, $\hat{p}_x = 0.983 \Rightarrow \hat{P}^* = 0.578 > \hat{P} = 0.576$; $PT = -1.28$, one-sided $p = 0.90$
- PT conditions on the marginal frequencies automatically: no need to guess the baseline
- Overlapping 5-day labels: PT assumes independent days; the same null as the slope t -test in $y_t = a + b x_t + u_t$ with Newey–West errors (Newey & West (1987)): $t = -1.00$
 - ▶ a : intercept; b : slope, zero under independence; u_t : error

 MFM_ch13_inference

Appendix: ROC AUC and Its Sampling Uncertainty

- Receiver operating characteristic (ROC) curve: true positive rate $TP/(TP + FN)$ against false positive rate $FP/(FP + TN)$ over all thresholds
- Area under the curve (AUC) $= P(\hat{p}_{\text{pos}} > \hat{p}_{\text{neg}}) + \frac{1}{2}P(\hat{p}_{\text{pos}} = \hat{p}_{\text{neg}}) = U/(n_1 n_0)$
 - ▶ $\hat{p}_{\text{pos}}, \hat{p}_{\text{neg}}$: predicted probabilities of a randomly drawn positive (up) and negative (down) observation
 - ▶ n_1, n_0 : numbers of positives and negatives; U : Mann–Whitney statistic, the number of (positive, negative) pairs ranked correctly, ties counted $\frac{1}{2}$
 - ▶ 0.5 = no ranking ability; 1 = every positive ranked above every negative
- DeLong standard error (DeLong, DeLong & Clarke-Pearson (1988)): from the placement values, the share of the other class that each observation outranks; assumes independent observations
- Circular block bootstrap (Künsch (1989)): resample blocks of $L = 20$ consecutive days, 1,000 replications; keeps serial dependence
- Logit, pooled out-of-fold AUC = 0.479
 - ▶ DeLong 95% CI [0.465; 0.494]: “significantly worse than random”
 - ▶ block bootstrap 95% CI [0.455; 0.504]: noise around 0.5
- An AUC below 0.5 is anti-ranking only if it survives a test that respects the dependence

Appendix: Giacomini–White, Statistic and Rolling Window (1/2)

- Giacomini & White (2006): the null concerns the forecasting **method** (model + estimation window + refitting rule), not the population model

$$H_0 : \mathbb{E}[d_{t+\tau} \mid \mathcal{F}_t] = 0 \Rightarrow \mathbb{E}[h_t d_{t+\tau}] = 0, \quad GW = n \bar{Z}^\top \hat{\Omega}^{-1} \bar{Z} \xrightarrow{d} \chi_q^2$$

- Notation

- ▶ $d_{t+\tau}$: loss differential of the two forecasts made at t for horizon $\tau = 5$ days; $\mathcal{F}_t$: information available at t
- ▶ h_t : test functions known at t (here 1 and the lagged loss differential, $q = 2$); $Z_t = h_t d_{t+\tau}$
- ▶ $\bar{Z}$: mean of Z_t over the n forecasts; $\hat{\Omega}$: HAC covariance matrix of Z_t ; χ_q^2 : chi-squared with q degrees of freedom

- Reading: under H_0 no information at t predicts which method will do better; a large GW (small p) rejects equal conditional accuracy

Appendix: Giacomini–White, Statistic and Rolling Window (2/2)

- The theory requires a bounded (rolling) estimation window, so that estimation noise never vanishes; an expanding window lies outside the theorem
- The question each test asks
 - ▶ CW: does the *population* model with features add information?
 - ▶ GW: does *my estimated* rule, as deployed, forecast better?
- S&P 500 logit, rolling 8-year window, as the GW theory requires: $GW = 26.5$ ($p < 0.001$): the logit forecasts worse than the constant

 MFM_ch13_inference

Appendix: Feature Importance: SHAP

Definition 7 (Shapley value of feature j (SHAP, Lundberg & Lee (2017)))

- Average marginal contribution of j over all coalitions $S \subseteq \{1, \dots, p\} \setminus \{j\}$:

$$\phi_j = \sum_S \frac{|S|! (p - |S| - 1)!}{p!} [v(S \cup \{j\}) - v(S)], \quad \hat{f}(x) = \phi_0 + \sum_j \phi_j$$

- Interventional SHAP: $v(S) = \mathbb{E}[\hat{f}(x_S, X_{\bar{S}})]$, averaging $\hat{f}$ with the features outside S drawn from a background sample (as in the Quantlet); $\phi_0 = v(\emptyset) =$ average prediction
- Notation: p : number of features; S : a coalition, a subset of the other features, $|S|$ its size, $\bar{S}$ the features outside it; $v(S)$: prediction when only the features in S are known; $\hat{f}(x)$: model prediction for x
- The weight of S is the probability that exactly the features in S come before j in a random ordering of the p features
- Conditional SHAP uses $v(S) = \mathbb{E}[\hat{f}(X) \mid X_S = x_S]$; the two differ when features are dependent
- Additive attribution of each prediction; mean $|\phi_j|$ over observations = global importance
- **Substitution effects**: correlated features share (and dilute) importance — cluster them first

Appendix: Reading Feature Importances

- **MDI** ranks volatility features first; **MDA** and **SHAP** point to 20-day momentum and the distance from the 200-day moving average
- 20-day volatility: high MDI but **negative** MDA — a sign of overfitting, not a proof: sampling noise and correlated features also move MDA
- Importances are tiny ($\Delta\text{AUC} < 0.01$), consistent with the model comparison
- Permuting one of several correlated features creates points off the data manifold: MDA and SHAP then **extrapolate** ([Hooker, Mentch & Zhou \(2021\)](#)); permute within clusters or refit without the feature
- Importances are estimates and need their own uncertainty analysis; honest trees give asymptotically Normal forest *predictions* and CIs ([Wager & Athey \(2018\)](#)), not CIs for importances
- **Backtesting is not a research tool**, importance analysis is: [López de Prado \(2018\)](#)

Appendix: Data-Snooping Tests: Reality Check, SPA, Romano–Wolf

- K rules, performance differentials $d_{k,t} = r_{k,t} - r_{0,t}$ against a benchmark r_0 ; $H_0 : \max_k \mathbb{E}[d_{k,t}] \leq 0$
 - ▶ n : number of days; $\bar{d}_k$: mean differential of rule k
- Reality Check (RC, [White \(2000\)](#)): statistic $\max_k \sqrt{n} \bar{d}_k$; null distribution from the **stationary bootstrap** ([Politis & Romano \(1994\)](#)) of the whole $n \times K$ matrix
 - ▶ resampling whole days keeps the correlation between rules: no need to count “effective” trials
- Superior Predictive Ability (SPA, [Hansen \(2005\)](#)): studentised $\max_k \sqrt{n} \bar{d}_k / \hat{\omega}_k$, $\hat{\omega}_k$ the standard deviation of $\sqrt{n} \bar{d}_k$; clearly bad rules are not recentred, so adding junk rules does not destroy power
- Romano–Wolf StepM ([Romano & Wolf \(2005\)](#)): step-down procedure that names *which* rules beat the benchmark, with family-wise error rate (FWER) control
- DSR asks “is the best SR above the luck of N trials?” and needs N and $\text{Var}[\widehat{SR}]$; these tests take both from the data

Appendix: Double Selection and DML, Penalty and Conditions (1/2)

- Partially linear model: $\mathbb{E}[\varepsilon_t \mid d_t, x_t] = 0$, $\mathbb{E}[v_t \mid x_t] = 0$, $\text{Var}(v_t) > 0$; g , m approximately sparse
 - ▶ the errors are unpredictable from the regressors; d_t has variation of its own beyond the controls
- Post-double selection: the heteroskedasticity-robust standard error is valid under independent sampling, regularity conditions and sparsity $s^2 \log^2(p \vee n)/n \rightarrow 0$ (Belloni, Chernozhukov & Hansen (2014)); with time series, HAC errors are an approximation
 - ▶ s : number of controls that matter; p : number of controls; $p \vee n = \max(p, n)$
- LASSO objective, with feature-specific penalty loadings

$$\frac{1}{n} \sum_t (y_t - x_t^\top \beta)^2 + \frac{\lambda}{n} \sum_j \hat{\psi}_j |\beta_j|, \quad \lambda = 2c\sqrt{n} \Phi^{-1}(1 - \gamma/(2p))$$

- ▶ plug-in constants $c = 1.1$, $\gamma = 0.1/\log n$; loadings $\hat{\psi}_j^2 = \frac{1}{n} \sum_t x_{tj}^2 \hat{e}_t^2$, $\hat{e}_t$ the residuals of the previous step, iterated

Appendix: Double Selection and DML, Penalty and Conditions (2/2)

- DML (Chernozhukov et al. (2018)): with $\ell(x) = \mathbb{E}[y | x]$, it suffices (with consistency and regularity) that

$$\|\hat{m} - m\| (\|\hat{\ell} - \ell\| + \|\hat{m} - m\|) = o_P(n^{-1/2})$$

- ▶ $\|\hat{m} - m\|$: root mean squared estimation error of the learner for m ; same for ℓ
- ▶ $o_P(n^{-1/2})$: tends to zero in probability faster than $1/\sqrt{n}$
- ▶ e.g. both errors $o_P(n^{-1/4})$: each learner may converge more slowly than a parametric model, because only the product of the errors matters

Appendix: Machine Learning for Inference in Asset Pricing

- [Feng, Giglio & Xiu \(2020\)](#): does a new factor add pricing information given hundreds of existing ones?
 - ▶ double selection of the controlling factors from the zoo, then a valid t -test of the new factor's risk price
 - ▶ the inferential answer to [Harvey, Liu & Zhu \(2016\)](#), who raise the hurdle to $t > 3$
- Heterogeneous effects: honest causal forests give CIs for effects that vary with covariates ([Wager & Athey \(2018\)](#))
- Three rules
 - ▶ prediction and inference are different targets: a good predictor can give a biased $\hat{\theta}$
 - ▶ orthogonalise before you test, and cross-fit with folds that respect time
 - ▶ report CIs; “selected” or “important” is not “significant”

Appendix: Deep Learning for Sequences

Recurrent networks (LSTM, GRU)

- ▣ Hidden state carries memory: $h_t = g(h_{t-1}, x_t)$; g : a non-linear function with trained weights, x_t : input at t
- ▣ Gates control what is kept or forgotten, mitigating vanishing gradients
- ▣ Used for volatility, limit-order-book and intraday forecasting

Transformers

- ▣ Self-attention: $\text{softmax}(QK^\top / \sqrt{d_k})V$, with query, key, value matrices Q, K, V (linear maps of the inputs); d_k : key dimension; softmax turns each row of scores into weights that sum to 1
- ▣ Long-range dependence, parallel training
- ▣ Backbone of large language models ([Chapter 15](#))

Appendix: Time-Series Foundation Models

Pre-trained forecasters

- Pre-trained on large collections of series, used **zero-shot** (without re-training): Chronos (Ansari et al., 2024), TimesFM (Das et al., 2024), Moirai (Woo et al., 2024), Lag-Llama (Rasul et al., 2023)
 - Useful baselines for volatility and volume forecasting
 - For returns they inherit the same signal-to-noise limits
 - Pre-training data may overlap your test period: a new form of leakage
-
- Covered in depth, with the risk-forecast evaluation tools of [Chapter 8](#), in [Chapter 14](#)

Appendix: Reinforcement Learning, Generative Models, Explainable AI, Risk

Reinforcement learning (RL)

- ▣ A policy $\pi(a | s)$ chooses actions a in states s to maximise cumulative reward
- ▣ Optimal execution, market making, deep hedging: Buehler et al. (2019)

Generative models

- ▣ GANs, variational autoencoders (VAEs), diffusion models: synthetic scenarios for stress tests

Explainable AI

- ▣ Required by regulators (EU AI Act, credit scoring, model risk)
- ▣ SHAP, LIME (local interpretable model-agnostic explanations), partial dependence

ML for risk

- ▣ Value-at-Risk (VaR): conditional quantiles (quantile forests, quantile-loss boosting); Expected Shortfall (ES): average of fitted tail quantiles or a joint VaR-ES loss; volatility: a squared-return target (Chapters 7–8)

Appendix: Deep Learning in Asset Pricing, Data (Section III.A)

- Monthly returns of all CRSP stocks, January 1967 – December 2016; excess over the one-month Treasury-bill rate
- 46 firm characteristics in six groups: past returns, investment, profitability, intangibles, value, trading frictions (Table A.II)
 - ▶ Only stock-months with all 46 available: about 10,000 of 31,000 stocks; ranked each month into cross-sectional quantiles
- 178 macroeconomic series: 124 of FRED-MD ([McCracken & Ng \(2016\)](#)), the 46 monthly medians of the characteristics, 8 predictors of [Welch & Goyal \(2008\)](#); transformed to stationarity
- Split: training 1967–1986, validation 1987–1991, test 1992–2016

Appendix: Deep Learning in Asset Pricing, Method (Sections I–II, Appendix C) (1/2)

- Adversarial generalized method of moments: $\min_{\omega} \max_g \frac{1}{N} \sum_j (E[M_{t+1} R_{t+1,j}^e g(l_t, l_{t,j})])^2$ (eq. 3); g builds the test assets with the largest pricing errors, motivated by [Hansen & Jagannathan \(1997\)](#)
- Networks: an FFN for ω ; an LSTM that turns the macro history into 4 hidden states; the adversary g with 8 instruments; three steps: unconditional SDF, maximise over g , minimise again; the full model is labelled GAN (Section II.D)
- Tuning: 384 configurations ranked by the validation Sharpe ratio (SR); chosen: 2 layers of 64 units, dropout, Adam; ensemble of 9 fits (Table A.VIII); loadings: a second FFN fits $E_t[R_{t+1,i}^e F_{t+1}] \propto \beta_{t,i}$
- Benchmarks: the FFN of [Gu, Kelly & Xiu \(2020\)](#); the linear SDF $\omega_{t,i} = \theta^\top l_{t,i}$ on long and short characteristic legs, unregularised (LS) or elastic net (EN); FF3/FF5 tangency portfolios ([Fama & French \(2015\)](#))
 - ▶ $\omega_{t,i}$: weight of stock i ; $l_{t,i}$: its characteristics; θ : one coefficient per characteristic, the same for all stocks and months

Appendix: Deep Learning in Asset Pricing, Method (Sections I–II, Appendix C) (2/2)

- ▣ Metrics (II.F): the SR of F and two fit measures; $\hat{\epsilon}_{t+1,i}$: pricing error, the part of $R_{t+1,i}^e$ not explained by $\hat{\beta}_{t,i}F_{t+1}$
- ▣ Explained variation: one minus the average squared pricing error over the average squared excess return

$$EV = 1 - \frac{\overline{\hat{\epsilon}^2}}{(\overline{R^e})^2}$$

- ▶ overline: time average of the monthly cross-sectional means; $EV = 1$: the factor explains all variation
- ▣ Cross-sectional R^2 : how well the model explains average returns

$$XS-R^2 = 1 - \frac{\frac{1}{N} \sum_i \frac{T_i}{T} (\bar{\hat{\epsilon}}_i)^2}{\frac{1}{N} \sum_i \frac{T_i}{T} (\bar{R}_i^e)^2}$$

- ▶ $\bar{\hat{\epsilon}}_i$, $\bar{R}_i^e$: time averages for stock i ; T_i : months of data on stock i , T : months in the sample

Appendix: Deep Learning in Asset Pricing, Results (1/2)

- Source: [Chen, Pelger & Zhu \(2024\)](#), *Management Science*
- Monthly out-of-sample SR of the SDF (Table I): GAN 0.75, EN 0.50, FFN 0.44, LS 0.42; FF5 0.22, FF3 0.19 (Table A.VI)
 - ▶ Reading: annualised 2.6 vs. 1.7 (EN) and 1.5 (FFN); the regularised linear no-arbitrage model (EN) already beats the forecasting network; unregularised LS does not
- Individual stocks (Table I): EV 8% (GAN) vs. 3–4%; cross-sectional R^2 23% vs. 14–19%
 - ▶ Reading: the no-arbitrage objective improves both fits, although it maximises neither
- Deciles sorted on the loadings β (Table II, test period, annualised): top minus bottom 0.39; FF5 alpha 0.39, $t = 9.96$; the GRS test rejects the CAPM, FF3 and FF5 ($p = 0.00$)
 - ▶ alpha: average return not explained by the factor model; GRS (Gibbons–Ross–Shanken) test: H_0 all alphas of the decile portfolios are zero
 - ▶ Reading: loadings predict returns, linearly ($R^2 = 0.97$ of mean return on β , Figure 8), and the factor models do not price them

Appendix: Deep Learning in Asset Pricing, Results (2/2)

- 46 sets of characteristic-sorted deciles (Table III): GAN cross-sectional R^2 above 90% for all; EV short-term reversal 0.70 vs. 0.43 (EN), momentum 0.54 vs. 0.26
 - ▶ Reading: gains come from extreme deciles and interactions; size and book-to-market deciles are easy for every model (Section III.E)
- Macro states (Figure 6): the 178 raw series make the out-of-sample SR collapse; GAN loses about 10% without macro states and 20% with unconditional moments only
 - ▶ Reading: the dynamics of macro data must be modelled (LSTM), not fed as the last increment
- Largest stocks (Table IV, monthly SR, models fitted on all stocks, evaluated on the subset): about 1,500 stocks GAN 0.41, FFN 0.30, EN 0.15; about 550 stocks 0.26, 0.24, 0.23; re-estimated on the 1,500: 0.42, 0.28, 0.42
 - ▶ Reading: the edge narrows sharply among the largest stocks; EN catches up when re-estimated on them

Appendix: Deep Learning in Asset Pricing, Replication and Extension

- Data limitation: the stock-level panel (CRSP returns, 46 characteristics) is not in the course data, so the GAN on individual stocks cannot be reproduced
- With the [Kenneth French Data Library](#) and FRED-MD, on the published split
 - ▶ Replicable: FF3 and FF5 tangency weights estimated on 1967–1986 (footnote 23), monthly SR on 1992–2016 next to Table A.VI
 - ▶ Not replicable with the course data: the published LS and EN (Tables I and III) use rank-weighted legs of 46 characteristics and stock-level betas
 - ▶ Analogue on portfolio data, an extension rather than a replication: the linear SDF ($\hat{\theta}_{LS}$ in closed form, $\hat{\theta}_{EN}$ with the objective of Section II.F) on halves (deciles 1–5, 6–10) of the 15 characteristics that the library sorts into deciles; its levels are not comparable with Tables I and III

Appendix: Deep Learning in Asset Pricing, Limits and an Open Extension

- Limitations stated by the authors
 - ▶ The positivity of the SDF is not imposed (footnote 9); only stock-months with all characteristics, which mainly drops very small stocks (Section III.A)
 - ▶ Frictions (Section III.I, Figure 17): annual SR 1.73 without the smallest 40% of stocks, 2.07 without the widest 40% bid-ask spreads, 1.87 without the lowest 40% turnover; see also [Avramov, Cheng & Metzker \(2023\)](#)
 - ▶ In-sample metrics exceed the out-of-sample ones, a sign of some overfitting (Section III.C); the hidden states are not comparable across fits, so they cannot be averaged (Section II.E)
- Open extension: freeze the 1967–1991 design and evaluate 2017–2026, after the paper's sample
 - ▶ does the linear no-arbitrage SDF still beat FF5?
 - ▶ do LSTM states of FRED-MD add to it?

Appendix: Deep Learning in Asset Pricing, Reading the Two Charts

- GAN 0.75 vs. FF5 0.22: with Lo SEs 0.065 and 0.058, the SE of the difference is at most 0.12 whatever their correlation, so the gap exceeds 4 SEs (i.i.d. Normal returns); a paired test on aligned returns ([Ledoit & Wolf \(2008\)](#)) is sharper
- The fixed FF5 weights load mainly on profitability and investment: RMW 0.42, CMA 0.30 (weights scaled to $\sum |w| = 1$); FF3 puts 0.61 on HML; both combinations were chosen on 1967–1986
- After 2016 the fixed tangency portfolios have monthly Sharpe ratios far below the market's; with $T = 116$ months the SE is about 0.09: a warning, not yet a test

 MFM_ch13_sdf_replication

Appendix: The Discovery Loop, Step by Step (1/2)

- **Literature:** "Summarise the evidence on how much published return predictability survives out of sample and after publication: DOI or arXiv id, sample, headline decay estimate."
 - ▶ Resolve every identifier; Semantic Scholar, Google Scholar, Elicit for the search
- **Hypotheses:** "For a feature that predicts S&P 500 returns in 2001-2015, give competing explanations (risk premium, reversal, microstructure, data mining) and what each predicts for 2016-2026 and for other indices."
- **Data and code:** "Write Python that builds 34 technical features from the course's daily S&P 500 and VIX closes with information up to day t only, and reports Newey-West t -statistics for 1-, 5- and 20-day returns in 2001-2015."
 - ▶ Any assistant (Claude, ChatGPT, Gemini, GitHub Copilot); first reproduce the case study: "always up" 58.1%

Appendix: The Discovery Loop, Step by Step (2/2)

- **Robustness**, fixed in advance: "Write the pre-analysis plan: the full list of features and horizons, the holdout, the corrections (Holm, BH, max-t) and the result that would count as a failure."
 - ▶ Placebo: shifting the target circularly (250–3,000 days) breaks its alignment with the features but keeps each series' own dependence; a valid null for $\max_j |t_j|$ if the target is stationary and the shifts exceed its dependence range
- **Critique**: "Act as a referee: which of these significant features are one effect counted several times, and which are known anomalies rather than new ones?"
- **Report**: all M tests, all trials including the discarded ones, DSR and PBO for any strategy built on them

References (1/9)

- ▣ Ané, T., Geman, H. (2000). Order flow, transaction clock, and normality of asset returns. *Journal of Finance*, 55(5), 2259–2284.
- ▣ Ansari, A.F. et al. (2024). Chronos: learning the language of time series. *arXiv:2403.07815*.
- ▣ Avramov, D., Cheng, S., Metzker, L. (2023). Machine learning vs. economic restrictions: evidence from stock return predictability. *Management Science*, 69(5), 2587–2619.
- ▣ Bailey, D.H., Borwein, J., López de Prado, M., Zhu, Q.J. (2017). The probability of backtest overfitting. *Journal of Computational Finance*, 20(4), 39–69.
- ▣ Bailey, D.H., López de Prado, M. (2012). The Sharpe ratio efficient frontier. *Journal of Risk*, 15(2), 3–44.
- ▣ Bailey, D.H., López de Prado, M. (2014). The deflated Sharpe ratio: correcting for selection bias, backtest overfitting and non-normality. *Journal of Portfolio Management*, 40(5), 94–107.
- ▣ Będowska-Sójka, B., Wójcik, P., Pele, D.T. (2026). Early warning systems for cryptocurrency markets: predicting ‘zombie’ assets using machine learning. *North American Journal of Economics and Finance*, 81, 102543.

References (2/9)

- ▣ Belloni, A., Chernozhukov, V., Hansen, C. (2014). [Inference on treatment effects after selection among high-dimensional controls](#). *Review of Economic Studies*, 81(2), 608–650.
- ▣ Benjamini, Y., Hochberg, Y. (1995). [Controlling the false discovery rate: a practical and powerful approach to multiple testing](#). *Journal of the Royal Statistical Society B*, 57(1), 289–300.
- ▣ Bergmeir, C., Hyndman, R.J., Koo, B. (2018). [A note on the validity of cross-validation for evaluating autoregressive time series prediction](#). *Computational Statistics & Data Analysis*, 120, 70–83.
- ▣ Breiman, L. (2001). [Random forests](#). *Machine Learning*, 45(1), 5–32.
- ▣ Buehler, H., Gonon, L., Teichmann, J., Wood, B. (2019). [Deep hedging](#). *Quantitative Finance*, 19(8), 1271–1291.
- ▣ Campbell, J.Y., Thompson, S.B. (2008). [Predicting excess stock returns out of sample: can anything beat the historical average?](#) *Review of Financial Studies*, 21(4), 1509–1531.
- ▣ Campbell, M., Hoane, A.J., Hsu, F. (2002). [Deep Blue](#). *Artificial Intelligence*, 134(1–2), 57–83.

References (3/9)

- Chen, A.Y., Lopez-Lira, A., Zimmermann, T. (2022). Does peer-reviewed research help predict stock returns? *arXiv:2212.10317*.
- Chen, L., Pelger, M., Zhu, J. (2024). Deep learning in asset pricing. *Management Science*, 70(2), 714–750.
- Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., Newey, W., Robins, J. (2018). Double/debiased machine learning for treatment and structural parameters. *Econometrics Journal*, 21(1), C1–C68.
- Clark, P.K. (1973). A subordinated stochastic process model with finite variance for speculative prices. *Econometrica*, 41(1), 135–155.
- Clark, T.E., West, K.D. (2007). Approximately normal tests for equal predictive accuracy in nested models. *Journal of Econometrics*, 138(1), 291–311.
- Das, A., Kong, W., Sen, R., Zhou, Y. (2024). A decoder-only foundation model for time-series forecasting. *Proceedings of the 41st International Conference on Machine Learning (ICML)*, PMLR 235.
- DeLong, E.R., DeLong, D.M., Clarke-Pearson, D.L. (1988). Comparing the areas under two or more correlated receiver operating characteristic curves: a nonparametric approach. *Biometrics*, 44(3), 837–845.

References (4/9)

- ▣ Diebold, F.X., Mariano, R.S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263.
- ▣ Dixon, M.F., Halperin, I., Bilokon, P. (2020). *Machine Learning in Finance: From Theory to Practice*. Springer.
- ▣ Fama, E.F., French, K.R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1), 1–22.
- ▣ Feng, G., Giglio, S., Xiu, D. (2020). Taming the factor zoo: a test of new factors. *Journal of Finance*, 75(3), 1327–1370.
- ▣ Franke, J., Härdle, W.K., Hafner, C.M. (2019). *Statistics of Financial Markets: An Introduction*, 5th ed. Springer.
- ▣ Friedman, J.H. (2001). Greedy function approximation: a gradient boosting machine. *Annals of Statistics*, 29(5), 1189–1232.
- ▣ Giacomini, R., White, H. (2006). Tests of conditional predictive ability. *Econometrica*, 74(6), 1545–1578.
- ▣ Gu, S., Kelly, B., Xiu, D. (2020). Empirical asset pricing via machine learning. *Review of Financial Studies*, 33(5), 2223–2273.

References (5/9)

- Hansen, L.P., Jagannathan, R. (1997). Assessing specification errors in stochastic discount factor models. *Journal of Finance*, 52(2), 557–590.
- Hansen, P.R. (2005). A test for superior predictive ability. *Journal of Business & Economic Statistics*, 23(4), 365–380.
- Harvey, C.R., Liu, Y., Zhu, H. (2016). ...and the cross-section of expected returns. *Review of Financial Studies*, 29(1), 5–68.
- Hastie, T., Tibshirani, R., Friedman, J. (2009). *The Elements of Statistical Learning*. 2nd ed. Springer.
- Hoerl, A.E., Kennard, R.W. (1970). Ridge regression: biased estimation for nonorthogonal problems. *Technometrics*, 12(1), 55–67.
- Hooker, G., Mentch, L., Zhou, S. (2021). Unrestricted permutation forces extrapolation: variable importance requires at least one more model, or there is no free variable importance. *Statistics and Computing*, 31(6), 82.
- Hosking, J.R.M. (1981). Fractional differencing. *Biometrika*, 68(1), 165–176.
- Kelly, B., Malamud, S., Zhou, K. (2024). The virtue of complexity in return prediction. *Journal of Finance*, 79(1), 459–503.

References (6/9)

- Kelly, B., Xiu, D. (2023). *Financial machine learning*. *Foundations and Trends in Finance*, 13(3–4), 205–363.
- Künsch, H.R. (1989). *The jackknife and the bootstrap for general stationary observations*. *Annals of Statistics*, 17(3), 1217–1241.
- Ledoit, O., Wolf, M. (2008). *Robust performance hypothesis testing with the Sharpe ratio*. *Journal of Empirical Finance*, 15(5), 850–859.
- Lo, A.W. (2002). *The statistics of Sharpe ratios*. *Financial Analysts Journal*, 58(4), 36–52.
- Lo, A.W. (2004). *The adaptive markets hypothesis*. *Journal of Portfolio Management*, 30(5), 15–29.
- López de Prado, M. (2018). *Advances in Financial Machine Learning*. Wiley.
- López de Prado, M. (2020). *Machine Learning for Asset Managers*. Cambridge University Press.

References (7/9)

- ▣ Lundberg, S.M., Lee, S.-I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems*, 30.
- ▣ McCracken, M.W., Ng, S. (2016). FRED-MD: a monthly database for macroeconomic research. *Journal of Business & Economic Statistics*, 34(4), 574–589.
- ▣ Nagel, S. (2025). Seemingly virtuous complexity in return prediction. NBER Working Paper 34104.
- ▣ Newey, W.K., West, K.D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708.
- ▣ Novy-Marx, R., Velikov, M. (2025). AI-powered (finance) scholarship. NBER Working Paper 33363.
- ▣ Opdyke, J.D. (2007). Comparing Sharpe ratios: so where are the p-values? *Journal of Asset Management*, 8(5), 308–336.
- ▣ Pesaran, M.H., Timmermann, A. (1992). A simple nonparametric test of predictive performance. *Journal of Business & Economic Statistics*, 10(4), 461–465.
- ▣ Politis, D.N., Romano, J.P. (1994). The stationary bootstrap. *Journal of the American Statistical Association*, 89(428), 1303–1313.

References (8/9)

- ▣ Rasul, K. et al. (2023). Lag-Llama: towards foundation models for probabilistic time series forecasting. *arXiv:2310.08278*.
- ▣ Robinson, P.M. (1995). Gaussian semiparametric estimation of long range dependence. *Annals of Statistics*, 23(5), 1630–1661.
- ▣ Romano, J.P., Wolf, M. (2005). Stepwise multiple testing as formalized data snooping. *Econometrica*, 73(4), 1237–1282.
- ▣ Rosenblatt, F. (1958). The perceptron: a probabilistic model for information storage and organization in the brain. *Psychological Review*, 65(6), 386–408.
- ▣ Shimotsu, K. (2010). Exact local Whittle estimation of fractional integration with unknown mean and time trend. *Econometric Theory*, 26(2), 501–540.
- ▣ Shimotsu, K., Phillips, P.C.B. (2005). Exact local Whittle estimation of fractional integration. *Annals of Statistics*, 33(4), 1890–1933.
- ▣ Tibshirani, R. (1996). Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society B*, 58(1), 267–288.

References (9/9)

- ▣ Wager, S., Athey, S. (2018). Estimation and inference of heterogeneous treatment effects using random forests. *Journal of the American Statistical Association*, 113(523), 1228–1242.
- ▣ Welch, I., Goyal, A. (2008). A comprehensive look at the empirical performance of equity premium prediction. *Review of Financial Studies*, 21(4), 1455–1508.
- ▣ White, H. (2000). A reality check for data snooping. *Econometrica*, 68(5), 1097–1126.
- ▣ Woo, G. et al. (2024). Unified training of universal time series forecasting transformers. *arXiv:2402.02592*.
- ▣ Zou, H., Hastie, T. (2005). Regularization and variable selection via the elastic net. *Journal of the Royal Statistical Society B*, 67(2), 301–320.
- ▣ **Code and data:** github.com/danpele/MFM/tree/main/Quantlets/Ch_13